

Review article

Enhancing EV battery lifecycle management: Robotic disassembly, design for disassembly, and sustainable solutions

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ABSTRACT

This article explores the growing importance of robotic disassembly for electric vehicle (EV) batteries, a critical issue as global EV usage continues to rise. With more EVs on the road, managing end-of-life batteries poses significant environmental and technological challenges. This review provides an in-depth analysis of recent advances in robotic disassembly, focusing on automation, artificial intelligence, and human-robot collaboration to improve the safety and efficiency of the process. The study underscores the urgent need for sustainable battery management strategies to reduce environmental impacts. Key obstacles include the lack of standardized battery designs and the risks associated with handling hazardous battery components. Additionally, the review highlights the potential of design-for-disassembly principles, which could facilitate easier recycling and repurposing of battery materials. A major focus is the assessment of current robotic technologies, particularly how artificial intelligence can enhance precision, minimize human involvement in dangerous tasks, and increase operational efficiency. The study also addresses the role of human-robot collaboration in overcoming the limitations of fully automated systems to ensure safe and flexible disassembly. Unlike most previous research that centers on chemical recycling, this review focuses on the mechanical disassembly process, addressing a critical gap. The article concludes with recommendations for future research and policy, emphasizing the development of advanced robotic systems and the need for regulatory frameworks that support sustainable practices throughout the EV battery lifecycle. These steps are essential for fostering an infrastructure that promotes environmentally responsible battery disassembly and resource conservation.

1. Introduction

Road transport is one of the key drivers behind the global rise in CO₂ levels and other greenhouse gases [1–3]. Reducing emissions and energy consumption in this sector is crucial to address the challenges of climate change, including global warming. Electrifying vehicle drivetrains is a vital step toward decarbonization, as it significantly lowers local emissions from road vehicles. Governments around the world are actively promoting the adoption of EVs as a means to decrease transport-related

emissions and slow the pace of climate change [4–6]. EV sales have surged significantly, with over ten million passenger vehicles sold in 2022. This number is expected to increase twofold by 2025 [7,8].

Although electric drivetrains are more energy-efficient and eliminate tailpipe emissions compared to internal combustion engines, they still have environmental consequences. Specifically, the extraction of raw materials for batteries and the manufacturing process contribute to pollution [9]. Additionally, mining for these materials can harm the environment and pose health risks to nearby communities [10–12].

Abbreviations: 3R, Recycling, Remanufacturing, and Reuse; AI, Artificial Intelligence; AR, Augmented Reality; ATC, Automatic Tool Changer; ATEX, Potentially Explosive Atmospheres; BMS, Battery Management System; CAD, Computer-Aided Design; CO₂, Carbon Dioxide; DfD, Design for Disassembly; DNN, Deep Neural Networks; DSP, Disassembly Sequence Planning; DTs, Digital Twins; EC, European Commission; EoL, End-of-life; EVBs, Electric Vehicle Batteries; HCC, Human-Controlled Combustion; HCI, Human-Computer interaction; HPSO, Hybrid Particle Swarm Optimization; HRC, Human-robot Collaboration; HRCD, Human-robot Collaborative Disassembly; LiBs, Lithium-ion Batteries; LoA, Level of Automation; ML, Machine Learning; MV, Machine Vision; NN, Neural Network; NSGAI, Non-dominated Sorting Genetic Algorithm II; OCV, Open-Circuit Voltage; OEM, Original Equipment Manufacturers; OEM, Original Equipment Manufacturer; PHEV, Plug-in Hybrid Electric Vehicle; QR, Quick-Response; RFID, Radio Frequency Identification; ROS, Robot Operating System; RP, Robot Programming; RUL, Remaining Useful Life; SMEs, Small and Medium-sized Enterprises; SoH, State of Health; SoR, Status of Risk; SoX, State of X, (X = health or charge or energy); SSM, Speed and Separation Monitoring; TP, Task Planning; TRL, Technology Readiness Level; XR, Extended Reality; YOLO, You Only Look Once.

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To reduce the environmental impact of road transport electrification, recycling end-of-life batteries and reusing functional ones is essential. EVBs that are no longer suitable for vehicle use can be repurposed for energy storage in buildings, extending their lifecycle and offering both energy efficiency and environmental benefits. When EVBs are no longer viable for reuse, recycling them into raw materials for new battery production is key to maintaining sustainable battery manufacturing practices.

BESS are crucial for improving renewable energy integration by addressing the intermittency of sources like solar and wind. They store excess energy during peak production and release it when demand is high, ensuring a stable energy supply. In smart grid implementation, BESS enhance grid flexibility and stability by balancing supply and demand, reducing the need for backup fossil fuel plants, and improving energy efficiency [24–30].

For demand response coordination, BESS allow consumers to store energy and use it during peak periods, reducing grid strain and lowering costs. In EVs applications, BESS support vehicle-to-grid and vehicle-to-home systems, allowing EVs to act as mobile energy storage, stabilizing the grid and reducing reliance on traditional energy sources.

Disassembly is a critical component of recycling and remanufacturing used products [13]. For effective reuse or recycling, the various materials in an assembly must be disassembled and sorted. However, disassembly is labour-intensive and particularly challenging when it comes to automating the process for used EVBs. While automation is possible, manual disassembly poses significant safety risks to workers handling degraded, EoL batteries. These risks include exposure to toxic substances, fire hazards, electric arcs, and high-voltage currents. Additionally, manual disassembly is often economically unfeasible in regions with high labour costs [15]. To reduce these risks and lower recycling costs, partial automation of EVB disassembly and sorting has been suggested [16]. Current approaches typically incorporate MV for environmental data collection, robotic arms for disassembly tasks, and AI for process management.

Automating the disassembly of EVBs is complex due to variations in mechanical condition, physical structure, and assembly techniques. Unlike standardized industrial products, EVBs often exhibit a lack of uniformity, with even batteries from the same manufacturer presenting multiple design differences [14,17]. Traditional industrial robots are optimized for repetitive tasks in structured environments where objects are fixed, whereas the disassembly of EVBs is more unstructured. This process requires flexibility in adapting to the type, condition, and structural design of the battery. As a result, advanced control systems and machine vision are needed to dynamically adjust key parameters during disassembly. Moreover, EVBs are not originally designed for easy disassembly, which adds to the complexity. Semi-autonomous tasks, welded connections, and adhesive bonds, often used in the production of new EVBs, further intensify the challenges. Tasks such as handling flexible wiring and autonomously removing electrical connectors and welded cell plates are particularly difficult. Consequently, HRC is frequently necessary for the more intricate disassembly tasks, which rely on cognitive abilities, fine motor skills, and craftsmanship. While HRC addresses several technical challenges, ensuring safety remains an ongoing concern.

The dismantling process of EVBs is far more complex than that of LiBs found in smaller consumer electronics, such as smartphones, laptops, and tablets. EVBs are designed with much higher energy storage capacities, making them significantly larger and heavier than their consumer electronics counterparts. As a result, EVBs are typically housed within a robust, reinforced structural frame, which serves to support not only the battery modules but also the control systems responsible for battery management, cooling, and safety. While destructive methods are often used in the disassembly of consumer LiBs, non-destructive techniques are preferred for EVBs. The recently implemented EU battery regulation [19] seeks to enhance the sustainability of electric vehicles by promoting the use of recycled materials in battery

production and creating a system for the collection of end-of-life EVs. Additionally, the EC has suggested the introduction of a digital battery passport to authenticate EVBs and provide detailed technical information to support recycling, repurposing, and remanufacturing processes [20].

This paper examines the current strategies for automating EVB disassembly, with particular emphasis on identifying the challenges and gaps in robotic autonomy for tasks such as disassembly, testing, and sorting. Although much research has focused on the chemistry and material recycling of EVBs, there has been limited exploration of how robotics and automation can be applied to the disassembly process. Existing research on robotics in battery disassembly mainly revolves around robotic control, autonomy, and HRC [14–22].

Additionally, this paper extends the scope of recent research by providing a comprehensive analysis of the challenges and potential opportunities that arise in the automation of the remaining manual procedures involved in EVB disassembly. While advancements have been made in automating various stages of battery disassembly, certain complex tasks still require human intervention. The paper delves into these specific areas, identifying key obstacles to full automation and exploring technological innovations that could help overcome these barriers.

In addition to the technical aspects of automation, the paper also critically examines the broader regulatory context, particularly the evolving EU battery regulations and the concept of the battery passport. The battery passport is an emerging framework designed to enhance transparency, traceability, and sustainability in the lifecycle of batteries, from production to disposal. By analysing these regulations, the paper evaluates how they may influence future strategies for automating disassembly processes, ensuring compliance with sustainability and safety standards.

This study presents a novel contribution to EVB lifecycle management by shifting the focus from chemical recycling to robotic mechanical disassembly—an area critical to automation yet underexplored in existing literature. While prior research has primarily emphasized material recovery techniques, our work systematically evaluates the technological feasibility, challenges, and advancements in robotic automation, AI, and HRC for EVB disassembly. By developing a standardized disassembly sequence and assessing key enablers such as safety considerations, machine vision, machine learning, and robotic programming, this study provides a comprehensive framework for enhancing automation efficiency in battery handling.

A key original perspective introduced in this study is the emphasis on DfD principles, which can significantly improve compatibility between EVBs and robotic automation. Unlike previous works that broadly advocate automation as a necessity, this study identifies specific design constraints—such as fastener types, modularity, and sensor placement—and proposes practical engineering solutions that facilitate robotic disassembly. Our findings indicate that the lack of standardization in EVB designs remains a primary bottleneck, reinforcing the urgent need for industry-wide regulatory frameworks to support scalable, sustainable automation.

Furthermore, this study highlights the underexplored role of HRC in EVB disassembly. Given the limitations of current AI and robotic systems in fully automating complex disassembly tasks, a hybrid approach—where human oversight enhances robotic precision—emerges as a realistic and immediate solution. To move beyond qualitative discussions, this study provides a quantitative assessment of automation feasibility, evaluating factors such as disassembly accuracy, cycle time reduction, and operational safety improvements.

Finally, this study proposes concrete steps to accelerate robotic EVB disassembly, including:

- Advancing machine learning and sensor-based AI models to improve real-time adaptability in disassembly tasks.

- Developing standardized EVB architectures that align with automation-friendly designs.
- Introducing regulatory incentives to encourage manufacturers to adopt DfD principles, thereby increasing automation efficiency.
- Enhancing safety protocols through AI-driven predictive models to mitigate risks associated with hazardous battery components.

By synthesizing insights from existing literature with original analyses on automation feasibility, safety, and regulatory challenges, this study not only bridges research gaps but also provides actionable solutions for advancing robotic EVB disassembly. The findings lay the groundwork for scalable, intelligent robotic solutions capable of addressing the growing challenges of EV battery recycling and sustainability.

To address these inquiries, an extensive review of the relevant literature is conducted. The structure of this paper is as follows: Section 2 outlines the taxonomy and methodology employed in this review. Section 3 highlights the necessity for robotic disassembly and identifies prevalent challenges within this sector. Section 4 evaluates current methods for disassembling EVBs, pinpointing deficiencies in automation. Section 5 explores existing shortcomings in automated disassembly processes and proposes approaches to further automate these manual operations. Section 6 considers future directions for the automation of manual tasks in EVB disassembly. The paper is concluded in Section 7. This structured examination offers a comprehensive guide for bridging the automation gaps in EVB disassembly, thereby enriching both scholarly discussion and practical developments in the area.

2. Classification and research analysis methods

This section details the objectives of the study, along with the methodology adopted for collecting, evaluating, and reviewing the existing literature. The primary focus is to explore the prevalent methods and identify any shortcomings in the robotic disassembly of EVBs.

2.1. Classification

The central objective of this review is to offer a comprehensive insight into the current practices in robotic EVB disassembly. The classification encompasses both automated and manual processes involved in testing, disassembly, and sorting. Tasks that rely on manual intervention are highlighted as areas requiring improvement, with recommendations to apply successful strategies from other domains to address these challenges.

2.2. Literature collection and screening methodology

This research explores various methods for disassembling EVBs, including those utilizing artificial intelligence. To maintain a structured and impartial review, a three-phase screening process was employed. Initially, a keyword search was conducted using terms such as electric vehicle AND battery AND robot AND disassembly AND recycling.

The literature search was carried out across several databases, including IEEE Xplore, MDPI, Springer, and Elsevier (with access to both Scopus and ScienceDirect). This yielded 2401 results from Elsevier, five from IEEE Xplore, 24 from MDPI, and no relevant results from Springer.

It is important to note that the Scopus database from Elsevier provided access to a broader range of articles beyond the specific databases mentioned. The literature search included publications up until August 30, 2024. In the second phase, manual filtering was performed to eliminate studies that did not directly address (a) EVB disassembly or (b) robotic disassembly approaches. Through this process, 70 pertinent articles were identified and incorporated into the reference list.

In the third phase, 58 additional articles referenced within these selected papers were reviewed and added to the final bibliography. This

brought the total number of included studies to 117, with articles categorized based on their content. Details of this classification process can be found in the appendix. After thorough screening, 133 publications focusing on robotic EVB disassembly were finalized for inclusion in the review section.

The inclusion and classification process is depicted in Fig. 1, which shows the steps involved in identifying, tagging, and assessing the selected studies. As the field of robotic EVB disassembly is relatively new, most of the research is concentrated in journal articles, with fewer contributions from books. The rise in publications over the past three years reflects the growing interest in this promising area for future research.

Fig. 2 illustrates the number of publications from 2014 to 2023 across three categories: Book Section, Conference, and Journal. The publication trends indicate fluctuating output in the early years, with a significant increase in journal publications starting from 2021. In particular, 2021 and 2023 show a sharp rise in journal contributions, reflecting growing academic engagement. Conference papers see a moderate but steady contribution, while book sections remain relatively limited, peaking in 2022. The overall trend highlights a shift toward journal publications, suggesting a growing focus on peer-reviewed research dissemination.

3. Historical context of automated production in EVB manufacturing

This section provides an in-depth overview of EVB disassembly, emphasizing the need for non-destructive, automated techniques to promote sustainable manufacturing. While there are differences in EVB designs, they generally follow a pack-module-cell structure, where each pack is made up of modules, which in turn contain individual battery cells. This modular design is still evolving and can be categorized into three primary types based on the cell design: prismatic, pouch, and cylindrical [22]. The disassembly process occurs at various levels, starting from the pack and moving down to the modules and then individual cells [14].

Extensive research has been conducted on disassembly, particularly at the pack-to-module level [23], as this stage is crucial for enabling component reuse, recycling, or repurposing. Where modules are used, the disassembly process involves separating these from key EVB components like mounting frames, wiring, hoses, and printed circuit boards.

By integrating BESS into renewable energy, smart grids, demand response, and EV systems, energy efficiency is optimized, leading to reduced environmental pollution and lower greenhouse gas emissions [31–33].

End-of-life (EoL) management of EVs is often similar to the disposal practices of internal combustion engine vehicles. However, the infrastructure for dedicated EV dismantling and sorting facilities is currently underdeveloped. Existing vehicle dismantling networks primarily handle decommissioned or accident-damaged EVs, often using manual methods without specialized disassembly technologies or expertise.

Due to the gradual implementation of new battery regulations, significant changes in existing practices are not expected in the near term. SMEs involved in battery pack collection and disassembly often face limitations in investment and a reluctance to adopt automation or robotic technologies, which further complicates the automation of disassembly processes.

3.1. Hierarchical structure of battery disassembly

Even though EVBs do not adhere to a standardized design, a general disassembly hierarchy can still be established. Fig. 3 outlines the essential tasks and subtasks involved in the disassembly process of EVBs. Before dismantling a battery pack, a preliminary risk assessment is conducted to evaluate the SoR related to the battery. If a battery is identified as damaged or carries a high-risk rating, it is not disassembled

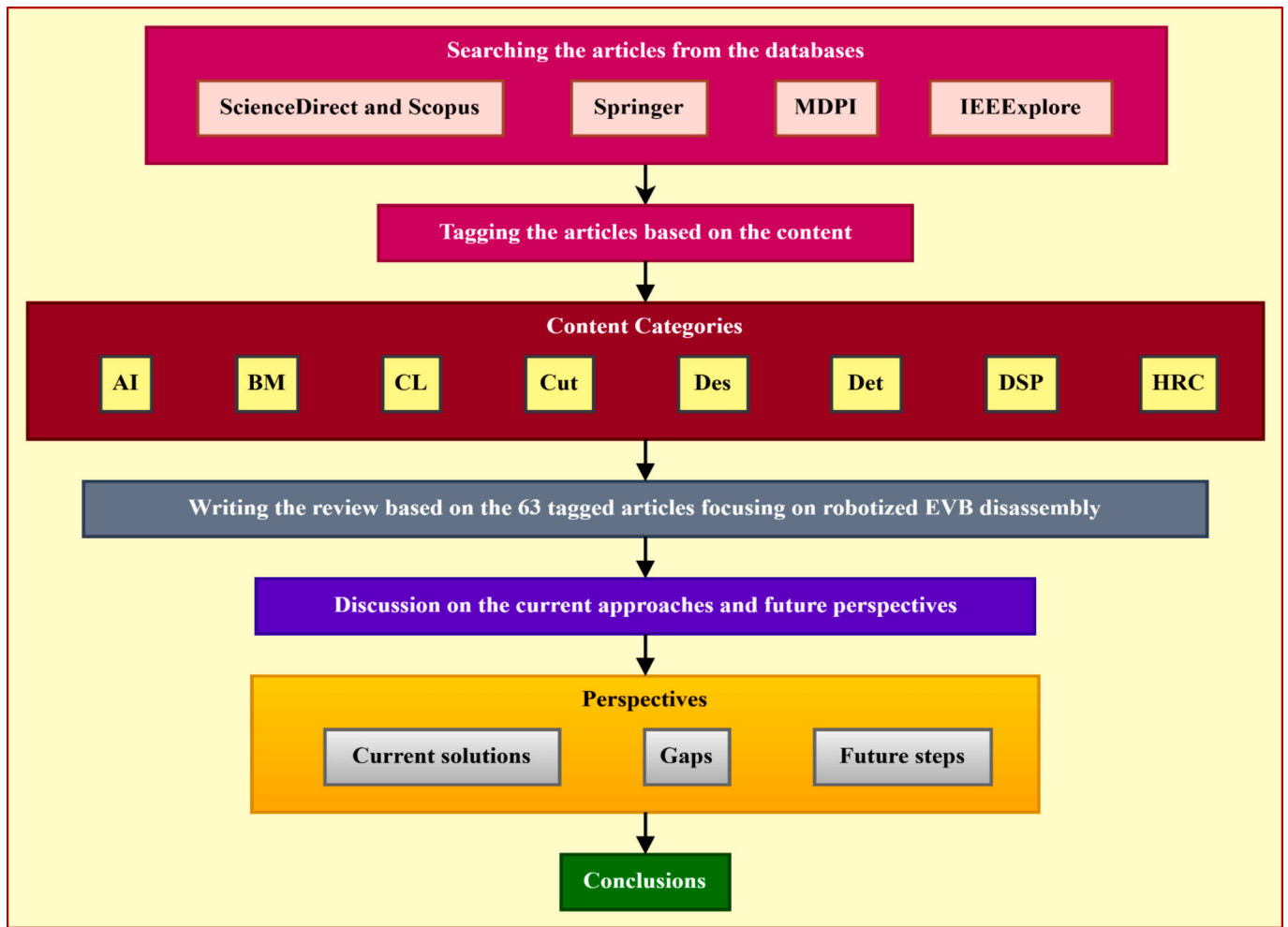


Fig. 1. Framework for searching, organizing, reviewing, and analysing EVB robotized processing.

for reuse, repurposing, or remanufacturing. Instead, such batteries are handled as whole units and sent directly for recycling through smelting methods [34].

For undamaged EVBs, two critical parameters are used to assess their potential for recovery: SoH and RUL.

The SoH reflects the current capacity of the battery in comparison to its original, optimal state [35]. Section 3.5 offers a comprehensive exploration of various techniques used to assess SoH and RUL. EVB modules that maintain a SoH above 80 % are regarded as suitable for ongoing application in vehicle systems [36]. This threshold ensures that the batteries are still in a good condition to perform effectively and reliably in automotive environments.

Once the SoH and RUL assessments are complete, disassembly progresses to the module level. Despite the variation in EVB designs, the fundamental steps involved in dismantling a battery pack to its individual cells are largely uniform [37,38]. These steps include removing components such as the battery cover, wire assemblies, bus bars, thermal management systems, and battery modules. In some EVB designs, thermal management systems may not be present, as they are not always necessary.

EVBs utilize a range of internal fastening techniques to ensure secure and effective assembly. These include threaded connections, which are commonly employed to fasten key components such as the battery cover, BMS, modules, and contactors. Threaded joints provide a strong and adjustable method of securing these elements, allowing for precise control over their alignment and tightness. In addition to threaded connections, welding is often used to create permanent, high-strength

bonds between metal parts, which is crucial for maintaining structural integrity and durability. This not only helps to hold these components in place but also improves thermal management by facilitating better heat transfer.

In some designs, reusable seals to prevent cover detachment replace adhesives. Flexible wiring assemblies are often secured using push-in rivets, while locking tabs are employed for electrical connectors. Additionally, retaining rings and push-in rivets are used to secure hose assemblies and quick fittings on hoses, respectively.

The method used to attach battery cells differs based on the cell type. For pouch cells, compression is achieved by stacking the cells and securing them with threaded bars at the corners along with metal end plates. Conduction plates are inserted between the cells and spot-welded to maintain connectivity. For cylindrical cell modules, spot-welded conduction plates are combined with a plastic housing to keep the cells together. Prismatic cells are firmly attached to the battery frame by means of threaded rods or screws, while bus bars are employed to interconnect the various individual modules.

The process of disassembling EVBs is highly intricate and involves several detailed steps. These steps typically include unscrewing components to dismantle the battery structure, cutting through materials to separate various sections, and carefully handling different parts of the battery, such as cells, wiring, and protective casings. Each of these components must be sorted for proper recycling or repurposing, with different materials—like metals, plastics, and chemicals—being managed separately to ensure environmental safety and regulatory compliance.

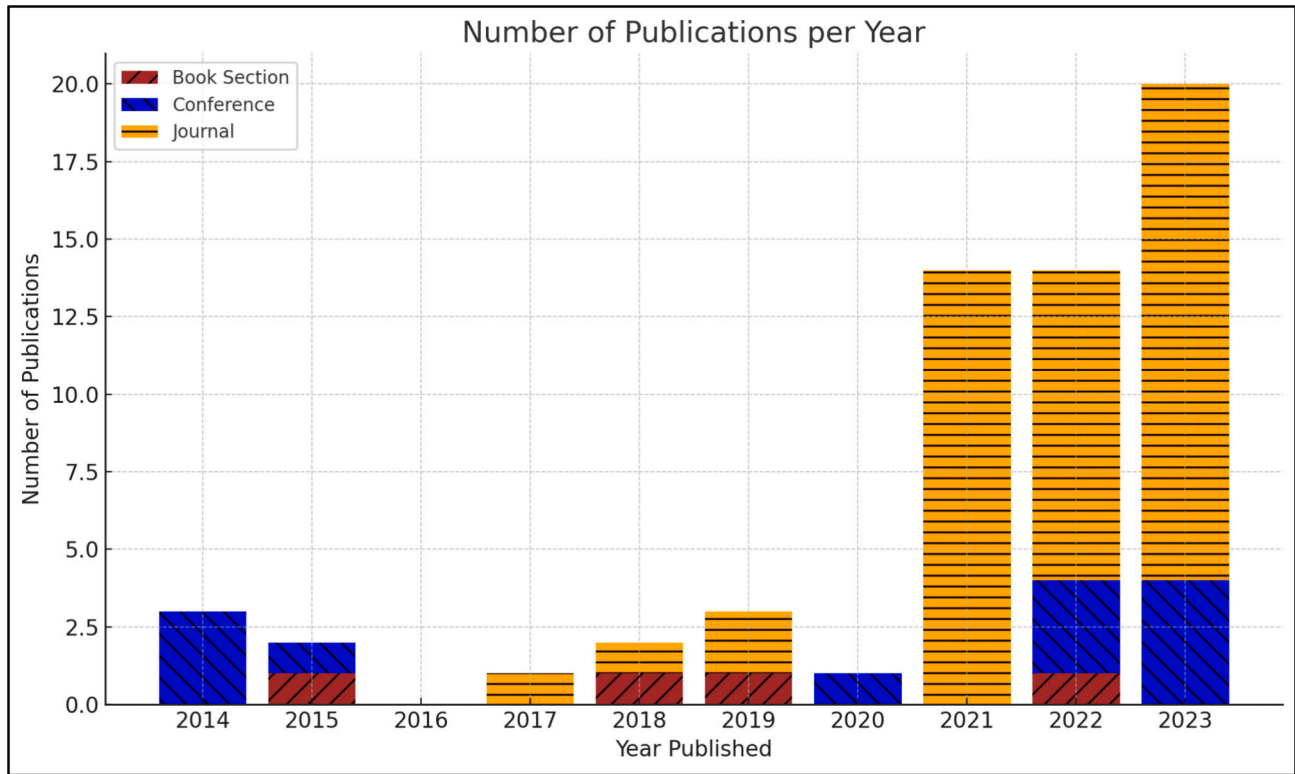


Fig. 2. Number of publications per year categorized by book section, conference, and journal.

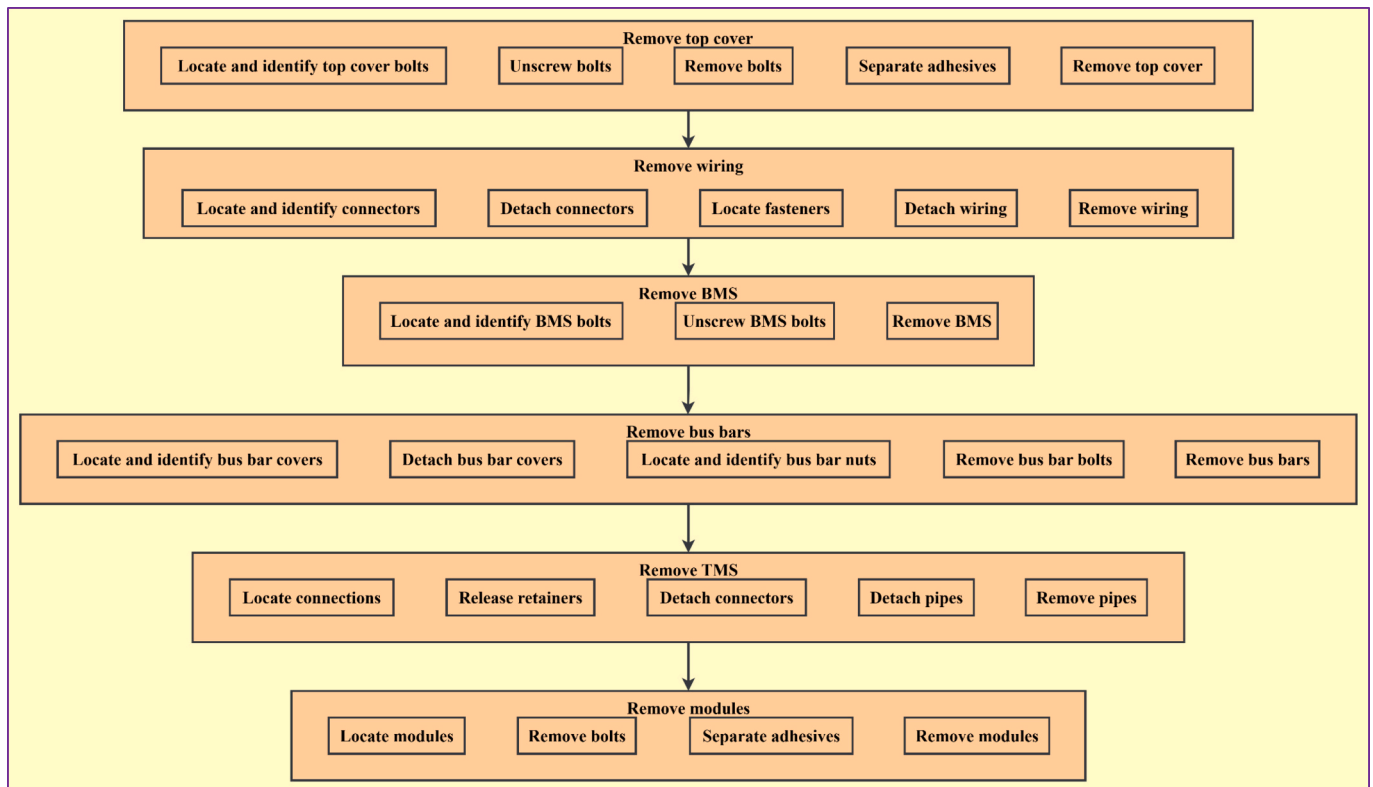


Fig. 3. The framework of EVB disassembly.

In addition to the physical disassembly, testing is a critical part of the process. This involves evaluating the SoH and the RUL of the battery. These tests provide insight into the battery's current performance levels

and its potential lifespan, which can guide decisions about whether the battery components should be recycled, refurbished, or reused. Accurate assessment of the SoH and RUL is essential for optimizing the recycling

process and minimizing waste.

A high-level task controller plays a critical role in managing the sequence of disassembly tasks. Through the integration of AI, these controllers can develop cognitive capabilities to identify components and enhance the efficiency of the disassembly process. Further details on the role of AI in this context are provided in Section 4.1.2.

3.2. Challenges in maintaining worker safety

A significant challenge in the disassembly of EVBs lies in ensuring the safety of workers, largely due to the intricate and potentially hazardous nature of the process. The diversity in EVB designs, caused by a lack of standardization, results in a wide range of physical characteristics that complicate the automation of the disassembly process. Furthermore, limited access to technical data concerning battery specifications, disassembly procedures, and chemical compositions adds another layer of difficulty. Harper et al. emphasize that manual disassembly requires highly trained personnel and stringent safety protocols, which subsequently increase operational costs [15].

The authors identified four major risk categories linked to LiBs, namely: electrical hazards, fire hazards, explosion risks, and chemical dangers. Electrical hazards are primarily due to the high voltage and substantial energy storage within the battery systems [39]. The risk of fires and explosions, on the other hand, is primarily attributed to the presence of flammable materials within the battery electrolytes or the hazardous by-products of various chemical reactions within the battery. These categories of risks are often interrelated, where the occurrence of one type of hazard can trigger others, resulting in a cascading series of potentially catastrophic events. Electrical hazards pose significant dangers during the disassembly process, as they can lead to electric shocks, short circuits, or thermal runaway reactions. These electrical failures may, in turn, result in the release of harmful gases, explosions, or fires.

To mitigate these risks, it is critical to incorporate automation into the EVB disassembly process. Automation not only enhances economic efficiency but also significantly improves worker safety by reducing direct human involvement in high-risk tasks. Disassembly systems need to be designed with ATEX compliance in mind, particularly when dealing with battery packs. The tools used in these systems should be capable of operating safely within ATEX zones when necessary. Manual disassembly processes also present a number of hazards, especially when workers are involved in disassembling and sorting battery components. The use of solvents to break down adhesive bonds between components can lead to the release of harmful fumes, thereby increasing the risk of fire. Similarly, the application of extremely high or low temperatures to separate adhesives poses risks such as frostbite or burns for workers. The use of cutting tools, like rotating blades, for separating welded joints adds another layer of risk, potentially leading to fires or causing eye injuries to workers. Implementing automation in these processes would significantly reduce the hazards, ensuring a safer and more efficient disassembly operation.

3.3. Lack of regulations and standards for batteries

A frequent challenge emphasized across various research studies is the significant diversity in battery pack designs, which poses major obstacles to automating disassembly processes. These design variations, including differences in size, shape, configuration, and assembly methods, create complexities for standardizing disassembly procedures. As a result, automation systems struggle to adapt to these variations, requiring either highly flexible machinery or manual intervention, which increases costs, time, and safety risks. This inconsistency makes it difficult to streamline disassembly, limiting the potential for widespread adoption of automated solutions in recycling or remanufacturing industries [40]. This variability means that each battery pack may require a different approach or set of tools for disassembly, making it difficult to standardize or streamline the process for automation. Thompson et al.

[41] emphasize that the diverse types of adhesives and fasteners used in battery packs, which further complicates automated disassembly efforts, exacerbate this problem. These variations require customized solutions for each battery type, adding to the complexity and cost of automation systems.

To mitigate this issue, Thompson et al. propose two primary design interventions aimed at facilitating disassembly automation: (i) the adoption of reversible adhesives that can be easily undone during the disassembly process without damaging the battery cells, and (ii) the shift from smaller battery modules to larger cells. These changes would not only simplify the disassembly process but also reduce the amount of material that needs to be managed during recycling or disposal. This is particularly important in the context of electric vehicles, where battery pack designs are often highly customized and vary significantly between manufacturers and models.

The authors [42] similarly emphasize the difficulties associated with non-detachable joints, such as those formed by welding or adhesive bonding. These joints often necessitate destructive disassembly methods, where parts of the battery pack must be broken or cut apart, which increases the potential risks associated with the process. Destructive disassembly can lead to hazards such as explosions, particularly when dealing with high-energy battery packs like those found in electric vehicles. This not only poses safety risks but also makes it harder to recover valuable materials from the batteries, further complicating recycling efforts.

Furthermore, Thompson et al. underscore the importance of standardizing both the design and labeling systems for battery packs, which would make it easier to automate the disassembly process. Currently, the lack of standardized battery designs means that automation systems need to be highly flexible, which drives up costs and reduces efficiency. The EU Battery Regulation has made some progress in addressing these challenges by mandating the labeling of batteries and introducing the concept of “battery passports” that would facilitate the sharing of essential information between manufacturers, recyclers, and regulators [43]. This regulatory framework aims to enhance transparency and improve the traceability of battery materials throughout their lifecycle.

However, despite these advancements, the absence of standardized battery pack designs continues to be a significant barrier to disassembly automation [44]. This challenge is further compounded by the varying conditions of EoL, which can differ significantly in terms of wear, degradation, or damage, adding complexity to the development of automated systems capable of efficiently handling diverse battery types and conditions.

The lack of stringent design regulations has resulted in a proliferation of different solutions for battery pack design, further complicating efforts to standardize and automate the disassembly process [45]. Manufacturers often prioritize performance and cost over recyclability and ease of disassembly, leading to the development of battery packs that are difficult to break down and recycle. As a result, the industry continues to face significant hurdles in its efforts to automate the disassembly process, which is a critical step in the sustainable management of battery materials and the development of a circular economy for electric vehicle batteries.

From February 18, 2025, batteries with a capacity of more than 2 kWh must include a carbon footprint report, detailing the manufacturer, model, production location, lifecycle carbon footprint (in kg per kWh), and a link to the full carbon footprint calculation. Additionally, these batteries must be labeled to indicate that their lifecycle carbon footprint complies with regulatory limits [19].

The regulation also promotes the use of recycled materials by mandating that manufacturers disclose the percentage of recycled raw materials in their batteries. This requirement will take effect on August 18, 2028, for batteries containing cobalt, lithium, or nickel. By August 18, 2031, minimum recycled content levels will be established: 16 % for cobalt, 85 % for lead, and 6 % each for lithium and nickel. These targets will rise to 26 % for cobalt, 85 % for lead, and 12 % for lithium and

nickel by August 18, 2036.

The objective of the regulation is to ensure that battery production meets sustainability goals while adhering to stringent environmental standards throughout the entire lifecycle of the battery. Manufacturers will be required to implement a take-back system for EVBs, ensuring that consumers can return batteries free of charge without the obligation to purchase a new one. This will apply regardless of the battery's brand, condition, or origin. The responsibility for collecting these batteries will also extend to businesses involved in recycling vehicles at the end of their lifecycle. Manufacturers will then transport the EVBs to appropriate facilities for processing and treatment.

Starting from February 18, 2027, all batteries will need to provide information on their general capacity and carbon footprint, which will be displayed directly on the battery. A QR code will be required on each battery, offering details about its composition, recycling methods, and a unique identifier linking to the upcoming digital battery passport system [20]. It will also allow stakeholders in the circular economy to trace a battery's lifecycle, from the sourcing of raw materials to final disposal, through an online platform designed for easy access.

The battery passport will make publicly accessible data available, including information on material composition, voltage, capacity, expected lifespan, operational limits, warranty, internal resistance, and end-of-life management strategies. Service and recycling professionals will have access to more technical specifications, such as material composition, part numbers, disassembly instructions, fastening techniques, necessary tools, safety protocols, SoX, potential risks, and the total number of cells in the battery [46].

Ford Motor Company, in collaboration with Everledger.io, is pioneering a battery passport system that will assign each EVB and its modules a unique identifier during production. This passport will include key data on the battery's chemistry, usage history, service records, updates, remanufacturing, and recycling processes. Ford's aim is to leverage the passport system to gain a comprehensive understanding of the entire lifecycle of EVBs, determining whether they can be recycled, repurposed for energy storage, or reused as remanufactured components [47]. The system will also provide detailed insights in their production, facilitating decisions on recycling, remanufacturing, or repurposing. Blockchain technology will be used to create a secure,

decentralized system for managing this information.

3.4. Evaluation of electric vehicle battery health and lifespan prediction

This section explores non-invasive testing techniques to assess the SoH and estimate the RUL of EVBs. We also examine various automated systems that can assist in the evaluation and testing of these batteries. Common approaches used to determine the condition of EVBs include direct assessments, statistical methods, and data-driven techniques. Fig. 4 illustrates a classification of the screening methods applicable to EVBs before disassembly.

Current research into EVB health estimation frequently underestimates the growing influence of ML and NN models. For example, the research discussed in [49] delves into various neural network architectures to improve the precision of predictions for SoH and RUL in batteries. This study emphasizes the growing importance and widespread adoption of ML techniques in battery condition monitoring. By refining the structure of neural networks, the researchers aim to enhance the accuracy of these predictive models, which play a crucial role in optimizing battery performance, prolonging battery life, and ensuring reliability in various applications. This reflects the broader trend of integrating advanced ML algorithms into battery management systems, underscoring the transformative potential of these technologies in managing and predicting battery health metrics.

Similarly, [50] examines the use of back-propagation neural networks for assessing the SoH of EVBs being repurposed for secondary use, thus showcasing the potential for extending the life cycle of these technologies. Another study [51] demonstrated how sophisticated ML frameworks can improve the accuracy of EVB health diagnostics. This research, along with others, highlights the merging of conventional diagnostic techniques with cutting-edge neural network approaches. Such integration leads to more precise predictions of the SoH and RUL of batteries, playing a vital role in the advancement of sustainable automotive technologies. This collective work contributes to the evolution of electric vehicle systems, promoting more efficient, reliable, and eco-friendly transportation solutions.

One notable technique for SoH estimation, particularly within certain specialized fields, is the deployment of DNNs, as detailed in [52].

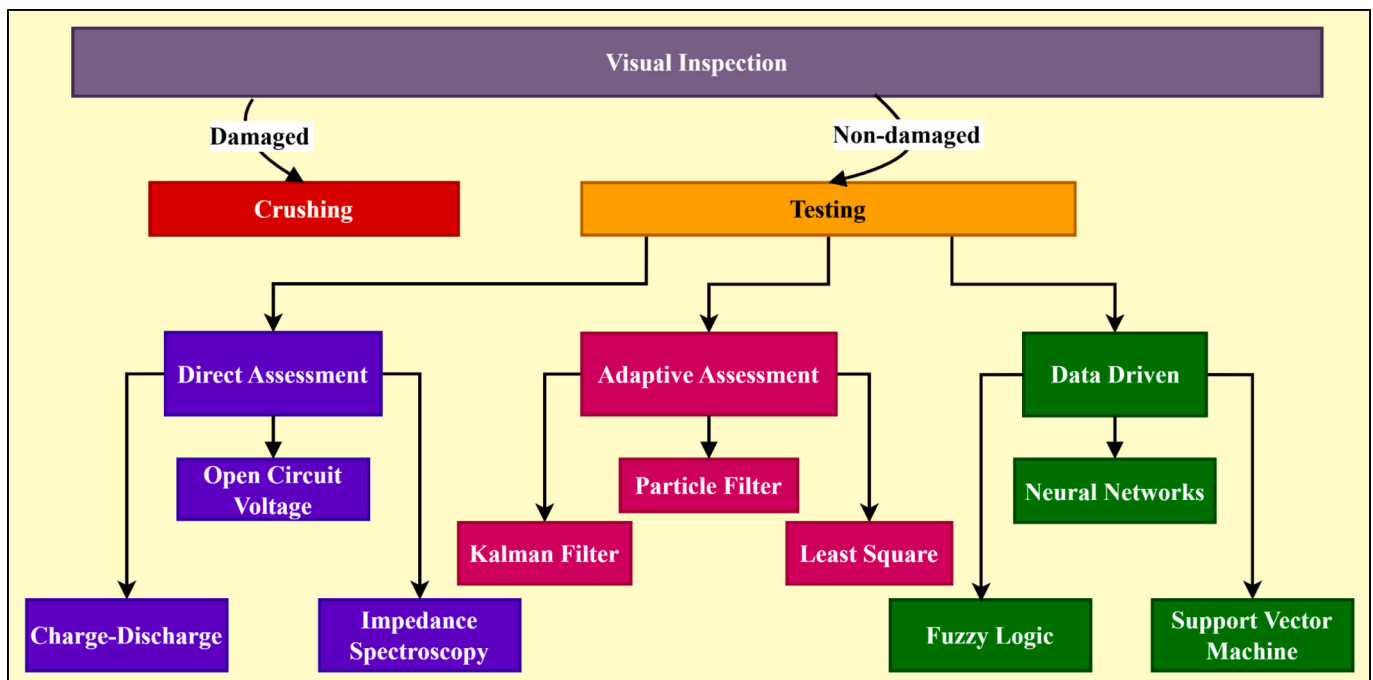


Fig. 4. Methods for monitoring of battery's health [48].

This approach uses a series of trained DNN models to deliver dependable health estimates by synthesizing outcomes from multiple networks, each developed from distinct data sources. This strategy is especially useful when working with data from various commercial LiB cells, which may differ in cathode materials and manufacturing processes, thereby allowing for uncertainties during the training process to be mitigated.

The implementation of neural networks has propelled advancements in EVB health evaluation, a field that continues to evolve at a rapid pace. Real-time applications of these models in battery diagnostics, as presented in [53], offer practical insights into both the challenges and potential solutions. Moreover, data-driven approaches focused on optimizing fast-charging protocols are increasingly leveraging advanced ML algorithms [54], emphasizing the essential role of AI in safeguarding and managing battery health. Furthermore, advanced techniques like are increasingly being studied for their potential in enhancing real-time health monitoring and online capacity estimation of LiBs. These techniques are particularly valuable due to their ability to process and analyze vast and intricate datasets efficiently, which is crucial in dynamic and real-time environments. In such applications, the rapid and accurate assessment of battery health and capacity is essential to ensure that performance remains optimal and that the lifespan of the battery is maximized. Ensemble learning combines multiple models to improve prediction accuracy, while random forest regression provides robustness in handling non-linear relationships within complex data. Together, these methods offer a comprehensive approach to diagnosing battery conditions, making them highly suited for applications where precise, timely decisions are necessary for the effective management and long-term durability of LiBs [49,55,56].

Additionally, advanced methods like Gaussian process regression and support vector machines are frequently employed to estimate the SoH of batteries, especially in electric vehicles. These techniques target specific operational parameters, such as partial charge states and incremental capacity curves, to deliver more precise and detailed evaluations of battery health. By examining these targeted data points, these models offer valuable insights into battery degradation patterns and performance capabilities, thereby enhancing the accuracy of SoH predictions.

Each technique brings its own exclusive approach to the estimation of SoH and RUL in EVBs. Gaussian process regression, for instance, leverages probabilistic modeling to capture uncertainties in the predictions, making it useful in cases where data is scarce or noisy. On the other hand, support vector machines apply classification and regression analysis, making them effective in separating different health states of the battery based on specific operational patterns [57,58].

4. Review of the relevant literature

Robotics and autonomous systems have gained recognition as a promising solution to enhance the disassembly of EoL products, especially by mitigating the safety risks and inefficiencies associated with manual disassembly, as highlighted in multiple studies [15,17,59–62]. Despite this potential, the growing automation in disassembly operations introduces challenges in managing unpredictable external factors, as noted in [62].

To address these uncertainties and ensure the necessary flexibility in automated disassembly processes, the integration of technologies such as HCI, AI, and advanced perception systems, or a combination thereof, is crucial [38].

Research conducted in [63] examined the depth of disassembly required to define the “stopping point” for either partial or sub-module disassembly, aiming to optimize both economic and environmental outcomes. Similarly, the system proposed by [64] evaluates the worth of recovered parts or recyclable materials by gathering data on the product, enabling the identification of the optimal disassembly level. Interviews with industry experts across various sectors within the EVB value chain [65] provided insights into the extent of automation

potential in the 3R framework. Among these, recycling was found to have the greatest automation potential, with an average of 74.2 %, largely because non-destructive disassembly is not always required. However, experts also agreed that fully automating disassembly remains impractical given the current designs and structures of batteries.

Typically, automated disassembly begins with gathering data on the device, followed by the acquisition of both 2D and 3D images. These steps are succeeded by the detection and identification of components, determining their locations, generating a disassembly plan, and ultimately removing the components. Each stage of this process draws upon various fields of research, including battery testing, ML, RP, DSP, TP, HRC, and safety engineering. The following sections classify the updated studies according to their respective themes, emphasizing the key contributions of each.

4.1. Cutting-edge robotics applications in EVB disassembly

The process of disassembling EVBs, whether carried out through fully autonomous or semi-autonomous systems, demands a level of robotic sophistication that surpasses what is typically required for the assembly of new products. This heightened complexity arises due to the intricate nature of disassembly tasks, which often involve handling components with varying conditions, ensuring safe extraction of delicate or hazardous materials, and adapting to the unique configurations of used EVBs. As a result, the robotics designed for this purpose must be equipped with more advanced sensory capabilities, higher precision, and greater flexibility compared to those used in standard manufacturing processes, where tasks tend to follow uniform, predictable patterns. This section explores methods for managing these uncertainties using sophisticated programming, disassembly sequence planning, and HRC.

4.1.1. Robot programming

Traditional approaches for programming industrial robots, such as offline programming or point-to-point instructions via a teach pendant (teach pendant is a handheld device used to program and control industrial robots), have proven effective in generating repetitive tasks for batch production, particularly when the production sequence is pre-defined, as seen in the assembly of new items. These methods, however, are not well-suited for disassembling EVBs, where higher-level control systems are needed to accommodate variations in products.

Robot programming for disassembly is particularly challenging due to the time required, creating a significant hurdle for the broader adoption of robotics in this domain. The diversity in battery pack designs further compounds the issue. Existing robot programming methods tend to rely on outdated techniques, without incorporating modern tools or languages. To address this challenge, researchers have explored innovative programming techniques to minimize programming time and simplify the process for non-expert users. For instance, the authors of [66] propose using a SpaceMouse, offering six degrees of freedom, as a substitute for the teach pendant in online programming, while [67] introduces an offline programming method that utilizes data from CAD models to streamline robot programming for component manipulation, significantly reducing the programming time required.

Task-oriented programming, while less common compared to robot-oriented programming, offers potential for wider adoption. It has been demonstrated that task-oriented programming can be made accessible to non-expert workers via an intuitive user interface, significantly reducing the required programming time, as described by [68]. This approach could pave the way for broader use of robotic systems among individuals without formal expertise in robotics.

4.1.2. Sequencing disassembly operations and task planning

Optimizing the sequence of disassembly operations is essential when dismantling complex devices like EVBs. Due to the complexity of components, the need for multiple disassembly stations further complicates

the sequencing process [69]. A notable study [70] addresses this challenge by proposing a disassembly matrix to outline relationships, priorities, and constraints among components. Another promising solution discussed in [44] suggests generating the disassembly sequence directly from CAD models, using the STEP data exchange format (ISO 10303-21) to encapsulate essential information needed for disassembly at the device's EoL. The authors [71] introduced a task planner designed specifically for EVB disassembly, where an industrial robot is guided by a three-dimensional camera system integrated into a sequential framework built on the ROS framework. The process begins by capturing eight images of the battery pack from predefined positions. A top-down approach is employed, determining the disassembly order based on overlapping components. The trajectory for disassembly is generated using real-time camera input, eliminating the need for a digital model of the battery pack. Although this approach was initially developed for a specific battery model (Audi A3), it can be adapted for other screw-joined battery packs. However, the system focuses on task planner design and path accuracy, without incorporating specific tools for the disassembly process.

The researchers [16] proposed a behavior-tree framework for managing mobile robots performing tasks such as reassembling separated sections of EVBs. This framework is built around three main components: navigation, object pose tracking, and gripping, all categorized into distinct modules. Regarding task delegation between humans and robots, Hellmuth et al. [72] introduced two indices for decision-making, taking into account various parameters related to disassembly, helping to determine which tasks are suitable for human-robot collaboration. The authors [73] developed a HPSO approach, integrated with Q-learning, to determine the optimal process of disassembly for effective HRC.

The authors [63] conducted an optimization for the cost aimed at identifying the optimal disassembly level for EVBs, balancing profitability and environmental impact. This model helps in determining disassembly strategies that maximize profits based on the condition of the battery modules while providing a valuable decision-making tool for users.

The researchers [61,74] presented an innovative self-learning system for disassembly planning that leverages NeuroSymbolic methods. This approach integrates task and motion planning to enhance the system's capability to autonomously determine efficient disassembly procedures. This system is designed to function in dynamic and unstructured environments by converting continuous visual data into discrete symbols using neural predicates and probabilistic learning. These symbols represent relevant states of the disassembly task, transforming the planning problem into one of finding optimal action plans in symbolic space.

In another study, an ontology-based disassembly planning approach was introduced, which incorporates partially destructive rules [75]. This ontology model aids in decision-making through case-based and rule-based reasoning. Additionally, it provides guidelines for selecting appropriate tools for components that cannot be dismantled non-destructively. The study also developed an iterative method for generating parallel disassembly sequences, ruling out unrealistic sequences to arrive at feasible solutions.

In [76,77], an advanced cloud-based architecture was proposed for managing robotic disassembly cells. This system is designed to handle high-level functions such as task scheduling, data collection, storage, and processing, all facilitated through a cloud infrastructure. The cloud-based setup aims to streamline and optimize disassembly operations by leveraging remote resources and centralized management.

Furthermore, [78] conducted experimental research on the disassembly of EVBs, with a focus on analysing time and cost factors. Although this study is not directly related to cloud-based systems, the insights gained could be valuable for developing and enhancing cloud-based disassembly processes. By incorporating this experimental data, future advancements in cloud-based disassembly solutions could benefit

from improved efficiency and cost-effectiveness.

The research outlined in [79] explores optimization strategies for disassembly through adaptive planning techniques. These methods could significantly improve cloud-based disassembly systems by fostering greater autonomy in decision-making processes. This trend underscores a movement toward data-driven and optimized disassembly practices. It suggests potential benefits from integrating experimental approaches with cloud technologies, particularly in the context of EVB recycling.

4.1.3. Robotic cell framework and automation framework

To tackle the complexities associated with diverse battery designs, several studies, such as the one by [80], have explored robotic cell layouts that employ multiple robots. In [81], researchers proposed an infrastructure centered around a data-driven robotic disassembly framework. This system is designed with intelligent agent principles, enabling adaptive, goal-oriented behavior. The framework includes interconnected modules capable of evaluating battery packs and employing AI-based algorithms to determine the appropriate EoL treatment for batteries. These decisions then guide subsequent disassembly procedures. A perception system, like a vision-based setup, gathers data and transmits it to the cloud, allowing the execution unit (e.g., a robotic arm) to access and act on this information. A user-friendly human-machine interface (HMI) further enhances the system's usability by presenting the collected data clearly.

4.1.4. Robotic tool design

The design of tools is critical when it comes to robotic disassembly, particularly in the context of battery disassembly [80]. Tools must be capable of operating safely in environments where explosive risks and high-voltage isolation are concerns. Thus, specialized tools are required, which expand upon the standard options available in the market [82]. In [83], the researchers highlighted four essential tasks that are critical to the disassembly process: handling, separation, clamping, and monitoring. Each of these tasks plays a pivotal role in ensuring an efficient and precise disassembly operation. To carry out these functions effectively, specialized tools tailored for robotic applications are required. These tools are integral to automating the process, allowing for enhanced accuracy, control, and efficiency in robotic disassembly systems. Handling pertains to the careful manipulation of components, separation involves the detachment of parts, clamping secures the pieces during operations, and monitoring ensures real-time oversight of the process.

Numerous studies have examined the disassembly process and suggested specific tools for the job. For instance, [61] developed a compliant pneumatic torque actuator to assist with bolt removal. This tool is equipped with a camera to detect fasteners, allowing the robot to avoid errors in depth perception and minimize the risk of screw damage. Additionally, [84,85] designed specialized devices for cutting covers and plates, as well as a two-fingered gripper to facilitate the quick manipulation of components. In HRC disassembly scenarios, [86] utilized a genetic algorithm to optimize tool selection for specific tasks.

4.1.5. HRC

Duan et al. [91] investigated the application of mixed reality technology to improve both the safety and efficiency of HRC. Their proposed framework incorporates a digital twin of the physical HRC cell, specifically designed for battery disassembly. The system utilizes a 3D camera to capture and continuously monitor the physical workspace, providing real-time tracking of the human worker's position, as well as the locations of various objects within the environment. This integration of digital and physical elements allows for enhanced coordination and situational awareness during disassembly tasks.

This data, processed through DSP, enables artificial intelligence to identify different stages of the disassembly process. Using AR headsets, workers can engage in a virtual disassembly of EVBs, visualizing each

step. In experimental trials, the researchers assessed various performance metrics, such as time efficiency, success rates, and collision avoidance, comparing these outcomes to a pre-programmed HRC system that lacked AI, perception, and mixed reality capabilities. Their findings demonstrated a reduction in idle times for both the human operator and the robot, as well as improved task completion rates.

Maximizing the efficiency of HRC requires strategic division of tasks, as this not only streamlines operations but also leads to significant cost reductions when compared to manual methods. In [73], researchers tackled the challenge of uneven task allocation by focusing on the concurrent disassembly of multiple batteries across different cells, ensuring that resources are utilized efficiently. An important aspect of this study was the consideration of risks such as the deformation of battery components, which could affect the integrity and safety of the disassembly process.

Building on this, Wu et al. [92] developed a comprehensive mathematical model aimed at optimizing several key parameters within human-robot disassembly lines. Their model specifically sought to reduce the number of workstations required, minimize idle time between tasks, decrease the dependence on human labor, and lower overall operational costs. By incorporating these variables, the model provides a more efficient and cost-effective approach to disassembly processes. To solve this multifaceted optimization problem, the researchers employed the NSGA-II algorithm, which is known for its effectiveness in handling complex, multi-objective optimization challenges, ensuring that all factors were balanced and addressed to achieve the best possible outcomes for the system.

Another approach, which combines task scheduling with multi-agent reinforcement learning and Q-learning-based 2D robot path optimization, is presented in [93] and focuses on battery module disassembly. Robotic safety standards play a critical role in these systems. ISO 10218-1 [95] and ISO 10218-2 [96] form the foundation of collaborative industrial robot safety, which has been further enhanced by ISO/TS 15066:2016 [97]. This standard provides guidelines for risk assessments and safety protocols, tailored to different operational scenarios. SSM, which is particularly relevant for tasks requiring frequent human intervention, ensures continuous operation by maintaining a safe distance. If this distance is breached, the robot halts automatically. The robot's speed adjusts dynamically depending on proximity to the human operator, enabling faster operations when safety constraints are met. SSM can be applied to traditional industrial robots, facilitating collaborative work without direct contact [98]. However, its full implementation is currently restricted by the lack of validated real-time tracking sensors.

The upcoming revision of the ISO 10218 standard will integrate aspects of ISO/TS 15066 to create a unified safety standard for both industrial and collaborative robots.

4.2. Disassembly of EVBs utilizing ML and vision systems

An in-depth analysis presented by [45] highlighted the significant role AI and ML can play in automating various stages of EVB disassembly. These stages include sorting, testing, safety assessment, decision-making processes, target identification via MV, component separation, and overall handling. While these technologies hold great promise, a key challenge lies in gathering sufficient data to effectively train AI models. Evaluating the current solutions using the TRL framework, the study determined that the highest TRL reached thus far in the literature is level five, indicating that most solutions have been demonstrated and validated in laboratory settings.

In terms of machine vision applications for EVB disassembly, the primary focus has been on object localization and classification. For instance, Deng et al. [100] explored deep learning techniques to optimize camera exposure for screw removal in EVBs. Similarly, Quan et al. [101] proposed a two-step method for detecting screws during disassembly. In the first step, a trained regression model identifies regions

likely to contain screws, and in the second step, these regions are further examined to remove false positives. This approach demonstrated superior accuracy compared to existing models when tested on a similar number of images.

Another investigation by Al Assadi et al. [102] examined the use of machine learning for categorizing damaged and undamaged screws, including Torx, Phillips, and hexagonal types. The study focused on the unscrewing process, using CNN and ML-based approaches to compare the actual torque curve of the screwdriver with standard reference curves. Data was collected from industrial screwdrivers equipped with sensors for torque and speed, resulting in a dataset of over 1000 torque-angle curves. The system successfully detected the unscrewing process with high reliability.

In a related study, the researchers [103] designed a robotic system with a two-finger gripper for holding and unscrewing hexagonal head bolts. The robot scanned the EVB surface to locate screws, using force sensors to provide feedback during the process. This system effectively demonstrated how robotics could enhance automation in EVB disassembly.

Given the variety of screw types, automated classification systems are critical for selecting the appropriate tools. [61,104] implemented the integration of machine vision techniques with machine learning algorithms, such as YOLOv5, where a depth camera was used to identify fasteners. This approach combines a depth camera and algorithmic processing to improve detection accuracy. Similarly, in [105], an "Active Screw Detection" system was proposed, utilizing depth cameras and the YOLOX model for screw and bolt detection. This system uses a two-stage process where initial depth images identify approximate locations, and refined 2D and depth images captured from different angles improve localization accuracy.

Furthermore, [23] developed a method where an RGB point cloud is created using a depth camera. This method uses an instance segmentation network for pixel-wise classification, followed by precise component localization through comparison with a pre-existing point cloud in a defined coordinate system.

To address potential issues with robot trajectory deformations, Gerlitz et al. [42] employed 3D image registration techniques to compare scanned EVB modules with CAD models. Although model-based techniques may have limitations when dealing with flexible elements, [106] demonstrated the success of model-free point cloud semantic segmentation in identifying wires and other flexible components.

In another example from [23], X-ray fluorescence was utilized to analyze the material composition of EVB components using a handheld spectrometer. While effective in detecting metal alloys, the technique struggled with polymers and faced challenges related to long cycle times and surface coatings.

4.3. Improving EVB testing and disassembly procedures

Even with trained personnel, specialized equipment, and protective gear, the disassembly of EVBs remains a hazardous task. The chemical makeup, high voltage, and physical processes involved, such as cutting through adhesives, introduce various risks, including chemical exposure, electrical hazards, and fire. To enhance safety, it is essential to explore methods that remove human workers from direct interaction with the batteries during disassembly.

The use of robotics in EVB testing represents a notable advancement in this field. Robots, guided by visual servoing techniques, are being used for tasks like electrochemical impedance spectroscopy on lithium-ion batteries [107], which improves the precision of battery health assessments while increasing efficiency to meet growing industry demands. This integration of robotics exemplifies how automation is transforming battery testing and disassembly processes.

Rodriguez et al. [108] introduced a robotic system that relies on visual inspection to pre-sort damaged and undamaged battery modules, preventing defective modules from reaching human-operated voltage

testing stations. This system enhances worker safety and ensures that only safe modules are subjected to further human testing, such as OCV testing to assess the SoH of the batteries.

Additional researchers [59,109] have integrated testing capabilities into robotic grippers. Tan et al. developed a gripper with dynamic fingers that adjust to the size of the component, improving functionality.

A notable gap in the literature is the limited exploration of tele-robotics in EVB disassembly. The authors [110] investigated this area by presenting two methods for teleoperated disassembly and sorting. The first approach utilized a haptic control device with six degrees of freedom, controlling a robot arm with seven degrees of freedom. The second approach used identical robotic arms as master and slave devices. Both setups physically separated the human operator from the disassembly process, allowing them to control the robot remotely.

This study evaluated the success rates and completion times of various tasks, such as cutting, handling, sorting, and screwing. The findings indicated that using identical master-slave mapping significantly improved task completion times, while haptic feedback contributed to higher success rates. However, overall task completion times did not show significant improvement.

4.4. Battery design for disassembly

The principle of “Intelligent Design for Disassembly” [45,111], alongside progressive strategies such as design for remanufacturing and recycling, underscores the importance of using non-destructive and reversible methods for disassembly. This approach has been widely acknowledged by various researchers as essential in enhancing the efficiency and reliability of automated or robotic disassembly processes. A fundamental strategy for facilitating disassembly is the elimination of permanent joints like welding, adhesives, or glued connections [80]. Moreover, creating products that can be dismantled along the vertical axis [59] is identified as another pivotal factor. These design principles align with the goals of a circular economy, as they consider the product's EoL stage [112].

In their detailed techno-economic analysis, Lander et al. [111] examined the manual disassembly costs of six EVB packs available on the market, revealing substantial design variations across manufacturers that led to differences in disassembly costs. The authors proposed that the introduction of semi-automated (human-robot collaboration) or fully automated disassembly systems could significantly cut these costs. However, several studies have pointed out that fully automated disassembly is currently impractical due to the high degree of variation in battery designs [15,61,65,73,80,87–90,92–94,109], despite the fact that most batteries follow similar disassembly steps [17,61,63,82,85]. DfD remains a favored approach for reducing disassembly expenses and boosting the role of robotics and automation in future processes [45,65,111]. Although non-destructive disassembly methods are ideal for reuse and remanufacturing, many existing battery designs rely on permanent joints like welds or glue [40], which necessitate destructive techniques to separate the components [15,75]. Over the last five years, DfD has been a major focus of research, with the majority of related literature published between 2022 and 2023.

4.5. Existing publications and reviews

Upon completing the literature review, numerous studies were identified, most of which concentrate on the chemical properties and recycling of materials in battery disassembly. However, only a couple of publications specifically focus on integrating automation and robotics into the disassembly of battery packs.

One publication [14] briefly outlines robotic disassembly techniques, while another [21] presents a thorough analysis of research related to applying robotics and automation to battery disassembly. This includes topics like robotic control, autonomy, and HRC. These studies encompass a variety of interdisciplinary areas that are essential for

automating battery disassembly, such as DSP, TP, HRC, AI, and MV. Moreover, [109] updates the framework for battery disassembly to reflect advancements in current technologies.

The authors in [15] conducted a comprehensive review of topics related to EoL management of batteries. They emphasized the role of robotic disassembly, while also discussing challenges like standardization and the scarcity of data on used batteries. In [81], the presented study aimed to assess the status of robotic disassembly in the wider framework of a circular economy, taking into account the incorporation of Industry 4.0 technology.

Despite increased interest in robotic disassembly, the focus has shifted toward HRC applications rather than fully automated systems. It is essential to recognize that disassembly processes differ from reverse assembly since most devices are not built with disassembly in mind.

In the domain of robotic battery disassembly, the review [87] offers valuable insights into the significance of HRC in overcoming the challenges associated with battery disassembly. It delves into essential safety protocols and collaborative operational methods, which are critical for designing robotic systems that prioritize both safety and efficiency during battery handling. Furthermore, the review discusses communication interfaces and pinpoints technological obstacles, outlining a pathway toward enhancing sustainable and dependable robotic solutions to boost the effectiveness of battery recycling operations.

A comprehensive review [45] investigates the integration of AI into the disassembly process of EVBs, highlighting its transformative potential in the field. The study underscores the pivotal role of machine learning techniques in not only optimizing the overall disassembly process but also in streamlining decision-making by identifying the most efficient approaches. Specifically, the review advocates for the use of reinforcement learning to develop optimal disassembly sequences, which is particularly important in handling the complex and variable nature of EVB structures. In addition to machine learning, the review delves into the critical role of machine vision systems. These systems are integral in accurately identifying and categorizing various components within the battery, an essential step that enhances the precision and adaptability of automated disassembly operations. By incorporating machine vision, the process becomes more flexible, capable of adjusting to the differing conditions and compositions found in batteries of varying designs and states of wear.

One study explored the use of a vacuum technology gripper for the handling and sorting of EVB components, specifically designed to extract battery modules from their packs. Another research effort proposed a two-finger gripper equipped with force feedback and 3D-printed polylactide fingers for disassembling modules from cells. However, this study highlighted some limitations: the force feedback system exhibited sluggish response times, and the polylactide fingers were found to be insufficiently durable for dependable handling. Additionally, the industrial robot employed had a payload capacity of only four kilograms, which was adequate for handling 3D-printed mockups but inadequate for managing the actual weight of battery modules.

Two of the reviewed studies [84,85] proposed cutting the tabs and fasteners that secure the modules, using a spinning wheel cutter to slice through 3D-printed plastic replicas of charging modules. These setups effectively addressed concerns such as the potential for spark generation during the cutting process, which could present risks of fire, explosion, and eye injury.

However, another paper [82] advised against cutting due to the risk of sparks. None of the studies investigated the complexities of cutting metals in environments where explosive gases may be present, such as in scenarios where thermal runaway leads to cell leakage.

Only one study [109] discussed the separation of bonded joints, recommending a tool driven by pneumatic pressure to detach adhesive joints holding the battery cover to the frame. This approach, while still conceptual, relied on the lateral force generated by a pneumatic cylinder and mechanical advantage. However, the method has yet to be empirically tested. In addition to force application, freezing the adhesive to

below -115°C has been suggested as a way to reduce the force required for separation [118]. Further investigation is needed to fully understand the implications of this technique.

The review detailed in [113] underscores the pivotal role of de-manufacturing within the framework of a circular economy. It emphasizes the need to embed circular economy principles right from the initial stages of product design. This focus is particularly crucial in the context of developing robotic systems intended for the disassembly of batteries. The review advocates for a comprehensive design approach where considerations for the product's end-of-life are integrated from the beginning. By adopting this approach, the process of disassembling batteries can be made more environmentally sustainable and efficient. This strategy not only aligns with the principles of the circular economy but also enhances the overall efficacy of recycling processes by ensuring that products are designed with their end-of-life dismantling in mind. Such integration could significantly advance the objectives of battery recycling within the circular economy paradigm.

Zhang et al. [114] conducted a comprehensive analysis of the regulatory landscape in China regarding the management and recycling of discarded EVBs, encompassing both central and local regulations. Their study employed a detailed two-dimensional analytical approach to evaluate the key policies affecting the entire industrial chain associated with EVBs. This chain includes several critical stages: collection, storage, transportation, testing, disassembly, and repurposing. Their analysis revealed that each stage of this chain is significantly influenced by regulations imposed by various authorities. For instance, regulations related to fire safety and technical standards play a crucial role in determining the requirements for storage infrastructure, ensuring that EVBs are stored safely and efficiently. The study also identified that regulatory focus is predominantly on aspects of battery usage, testing, sorting, and disassembly. In contrast, there is a notable lack of comprehensive regulations addressing the storage, collection, and transportation of EVBs, with only 10 % of existing policies covering these latter areas.

Another relevant study conducted by the authors of [112] explored circular business models applicable to EVBs and various electronic devices. The review emphasized that enhancing product design is crucial for the successful application of circular economy principles. This finding underscores the importance of designing products with circularity in mind to facilitate more effective recycling and reuse processes.

Furthermore, the literature review incorporated insights from

additional studies [115–117], which delved into topics such as battery chemistry, supply chain dynamics for components, and material recycling. Although these subjects are significant within the broader context of battery management and recycling, they do not directly contribute to the primary focus of the review discussed here. Table 1 summarizes the focus, key contributions, and challenges addressed in some notable research studies.

5. Discussion

This section explores the findings of the detailed review presented in Section 4. The papers focusing on the robotic disassembly of EVB are organized in Table 2, according to the primary focus of each study. Section 5.1 presents an overview and examination of automated EVB disassembly techniques, with a particular focus on the contribution of AI in advancing these processes. The following subsections address tasks still dependent on manual efforts and analyze the challenges associated with automating these activities.

5.1. Current automation status in EVB disassembly

The automation of EVB disassembly involves seven critical tasks: separation, testing, sorting, cutting, grasping, detecting, and unscrewing. These tasks are illustrated in Fig. 5, which provides a breakdown of the research focus on each specific disassembly activity. This figure offers valuable insights into the first research question: which aspects of disassembly and sorting have been automated? Among these tasks, the automation of unscrewing has received particular attention, with nine studies dedicated to this process. Unscrewing is a crucial step in EVB disassembly due to the prevalent use of screws and bolts as fasteners. The emphasis on automating unscrewing reflects its significance in the overall disassembly process, with 36 % of the tasks mechanized through methods that do not involve AI. This highlights the practical importance of unscrewing and the ongoing efforts to improve automation in this specific area.

Despite the extensive review of literature, none of the studies provided methods for detecting the pitch of screw threads, which is crucial for synchronizing the robot's linear motion with the screw's rotational speed during the unscrewing process. This gap points to a significant limitation in current research, as it has predominantly concentrated on screw identification and localization rather than on enhancing the

Table 1
Key studies on robotics and automation in battery disassembly.

Study	Focus	Key Contributions	Technologies/Topics Covered	Challenges/Insights
[14]	Robotic disassembly techniques	Overview of robotic disassembly techniques	N/A	Limited to a brief overview
[21]	Robotics and automation in battery disassembly	Comprehensive analysis of robotic control, autonomy, and HRC in battery disassembly	DSP, Telepresence, HRC, AI, Machine Vision	Broad interdisciplinary impact, emphasis on technological integration
[109]	Battery disassembly framework	Updated disassembly framework to align with current technology	Generic mechatronic device disassembly	Proposal for a human-robot hybrid disassembly workstation with adaptable tools
[15]	EoL management of batteries	Extensive review on robotic disassembly, challenges of standardization, and data limitations	Robotic disassembly, EoL management	Focus on standardization and data challenges in EoL battery management
[81]	Robotic disassembly within the circular economy	Assessment of current robotic disassembly, integration of Industry 4.0 technologies	Industry 4.0, HRC	Shift toward HRC, differentiation between disassembly and reverse assembly
[87]	Robotic battery dismantling	Key insights on HRC's role in battery disassembly	HRC, safety requirements, collaborative operational modes	Safety, communication interfaces, technological challenges
[45]	AI in EVB disassembly	Application of AI for process optimization, machine vision for battery identification	AI, Machine Vision, Reinforcement Learning	Process optimization, flexibility, addressing varying battery states
[113]	De-manufacturing in the circular economy	Integration of circular economy principles in design	Circular economy, de-manufacturing	Promotes holistic strategy for environmentally sustainable disassembly
[114]	Central and local regulations in China for EVB processing	Two-dimensional analysis of policies impacting EVB industrial chain	Collection, storage, transportation, testing, disassembly	Regulatory impact, emphasis on technology development, focus on policy gaps
[112]	Circular business models for EVBs and white goods	Analysis of literature on circular economy models, device design improvement	Circular economy, design improvement	Need for device design enhancement to implement circular economy models
[115–117]	Battery chemistry and material recycling	Focus on foundational aspects of battery recycling, component supply chains	Battery chemistry, material recycling, supply chain	Not directly relevant to robotic disassembly but foundational for recycling processes

Table 2
The role of AI in enhancing robotics for recycling and safety applications.

Ref.	AI	Business Model	Cell Layout	Cutting	Design	Detection	DSP	Framework	Grasp	HRC	Legislation	Machine Vision	Unscrew	Recycling	Review	Robot Programming	Safety	Separation	Sorting	Teleoperation	Testing
[16]	✓						✓	✓							✓				✓		
[17]	✓		✓							✓		✓	✓						✓		
[23]	✓					✓						✓								✓	
[45]	✓											✓			✓						
[61]	✓						✓					✓	✓								
[71]	✓					✓	✓					✓									
[74]	✓										✓					✓			✓		✓
[88]	✓						✓			✓											
[100]	✓											✓	✓								
[101]	✓					✓						✓									
[102]	✓					✓						✓	✓								
[103]						✓						✓									
[104]	✓										✓										✓
[105]	✓					✓						✓	✓								
[106]	✓											✓									
[63]		✓					✓														
[111]		✓			✓											✓			✓		
[70]			✓							✓									✓		
[80]			✓		✓												✓				
[83]			✓											✓							✓
[98]			✓							✓		✓	✓			✓	✓				
[84]				✓					✓									✓			
[85]				✓					✓												
[15]					✓										✓				✓		
[40]					✓									✓							
[44]					✓		✓							✓			✓				
[60]					✓									✓							
[65]					✓									✓			✓		✓		
[84]				✓					✓					✓							✓
[85]				✓					✓					✓							

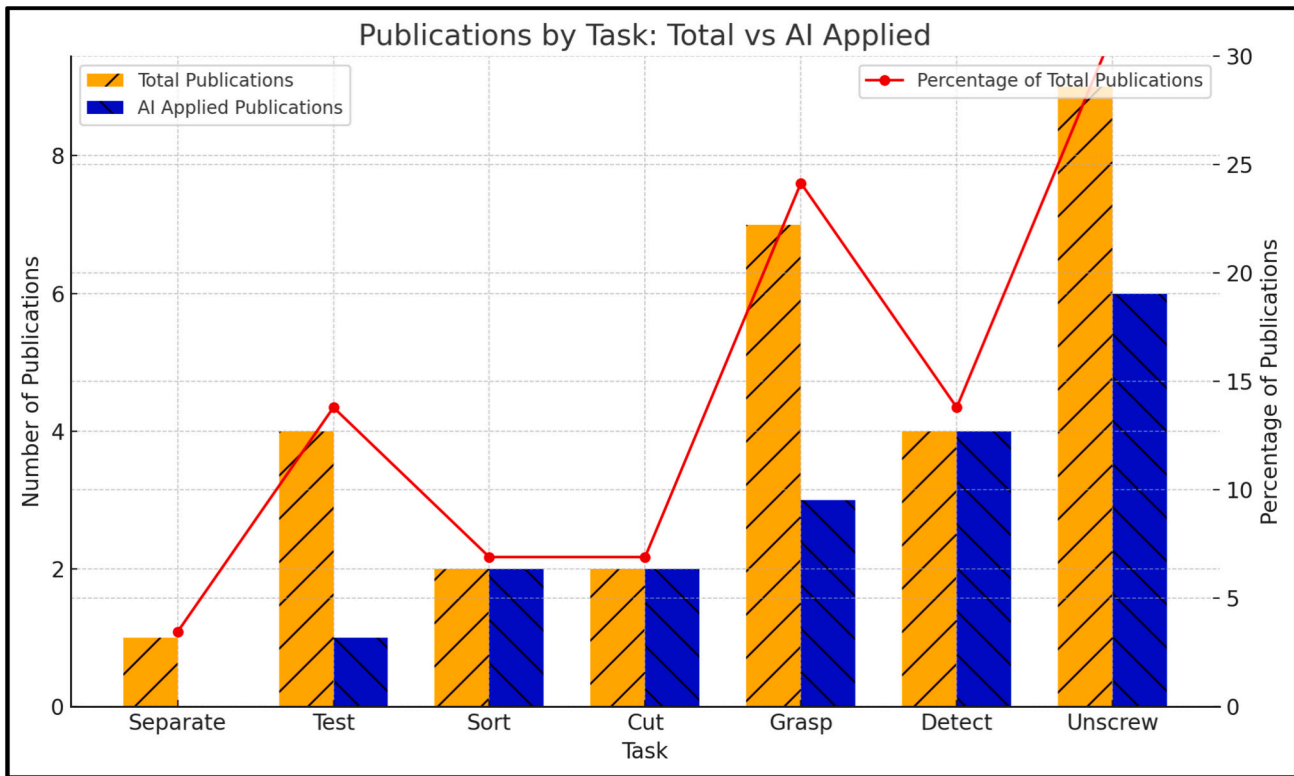


Fig. 5. Summary of robotised tasks.

efficiency of the unscrewing process itself.

Machine vision has emerged as the primary technique for recognizing and locating EVB components, often combined with force-feedback techniques for surface exploration. Among the seven studies addressing detection methods, four focused on identifying screws and their precise locations. Two studies examined the detection of other EVB components, including the BMS, modules, and mounting brackets. Additionally, one study employed machine vision to identify defective batteries prior to disassembly. Both two-dimensional and three-dimensional cameras were utilized in these detection processes, highlighting the diverse approaches to component recognition in EVB disassembly.

The variability in size and surface characteristics of EVB components poses significant challenges for robotic handling. This variability underscores the need for developing adaptive or multifunctional grippers that can accommodate different component specifications. Notably, only two studies have addressed the sorting of individual EVB components, indicating a gap in research related to this aspect of EVB disassembly.

Robotic testing and pre-sorting of damaged and intact battery modules can improve safety by ensuring that only undamaged batteries proceed through subsequent disassembly stages. Identifying and removing batteries that are leaking or structurally compromised is crucial, especially in scenarios where manual disassembly or HCC is involved, as these batteries pose significant fire and chemical hazards. One publication suggested a robotic visual inspection system for pre-sorting, while two other studies introduced a multifunctional gripper capable of directly assessing the OCV and internal resistance of EVB modules to determine their state-of-charge and safety status. Combining visual inspection with direct measurements could lead to the development of a robotic solution for pre-sorting and testing EVB modules, an essential step toward safe and efficient disassembly.

Manual processes, such as separating spot welds, disconnecting connectors, and removing electrical wire assemblies, continue to be predominant in the module-to-cell level of EVB disassembly. The

majority of research, over 90 %, has concentrated on automating the pack-to-module disassembly stage, leaving the automation of module-to-cell disassembly largely underexplored. This gap suggests a significant opportunity for advancing research in automating tasks like grabbing, cutting, sorting, separating, and testing at the module-to-cell level. Such advancements could pave the way for fully autonomous systems capable of managing both pack-to-module and module-to-cell disassembly processes.

Additionally, none of the reviewed studies addressed the integration of robotic disassembly with emerging battery passport technologies, such as the use of RFID tags or methods for identifying disassembled components to link them with their respective passports. This oversight indicates a need for research focused on combining robotic disassembly with battery passport systems to enhance traceability and management of EVB components.

5.2. Disassembly of the HRC

Numerous studies have explored the potential of HRC in disassembly processes. However, the term “collaborative” is often incorrectly applied, with “cooperative” being a more accurate description in many instances. Tasks labeled as collaborative are frequently cooperative in nature. Collaborative robotics involves simultaneous interaction between humans and robots in a shared workspace, where both actively contribute to the same task in real time. For example, a robot might hold a battery module while a human removes fasteners. Cooperative robotics, on the other hand, refers to sequential task execution, where the human and robot work toward a common goal but not necessarily at the same time or in direct interaction. For instance, a robot disassembles part of a battery pack, and then a human takes over to handle complex components.

Much of the research presents HRC as a strategy to address the variability in EVB disassembly. However, this typically results in humans taking on the more intricate tasks, such as handling connectors and manipulating flexible parts. While robotics technology has

advanced, the adaptability, human workers compared to disassembly robots still more efficiently handle flexibility, and dexterity required for these specific tasks.

For example, the removal of electrical connectors poses significant challenges due to high variability, making many automated solutions economically impractical. The diversity in connector types and configurations requires specialized tools, extensive expertise, and a range of disassembly techniques. Additionally, connectors are often located in tight spaces where surrounding components obstruct their visibility, reducing the effectiveness of optical sensors. On the other hand, human operators, equipped with cognitive skills, are able to efficiently manage these tasks, relying on decision-making abilities, dexterity, visual precision, and tactile feedback.

Some of the literature provides broad descriptions or preliminary investigations into potential disassembly workstations where humans and robots share the same workspace. Other studies focus on HRC primarily from the perspective of Dynamic Scheduling and Task Planning (DSP/TP). Only one study has applied HRC in practice to support human workers during the disassembly of battery modules, where robots assist by handling tools and securing parts. Often, HRC in disassembly is viewed as a complex case for testing DSP/TP algorithms in managing interactions with unpredictable agents, such as human workers. These studies frequently rely on simulations and theoretical models without validating their results through real-world experimentation.

The strong theoretical focus on HRC in most studies underlines the challenges of applying it in practical EVB disassembly, particularly due to safety concerns. While automating EVB disassembly can safeguard workers from hazardous procedures and lead to cost reductions, incorporating HRC may not necessarily improve safety when humans are directly involved in disassembling battery modules. In the reviewed literature, HRC is frequently proposed as a way to bypass the complexities of developing fully autonomous systems for complicated disassembly tasks.

5.3. The role of artificial intelligence and its challenges

AI has demonstrated significant potential in various stages of the battery disassembly process, as noted in numerous studies [45]. These stages include sorting, testing, safety oversight, decision-making, object recognition (such as identifying disassembly targets via machine vision), and the separation and handling of components. As shown in Fig. 5, AI has already been integrated into several disassembly tasks. For specific actions, such as sorting, cutting, and grasping, AI has been employed to enhance their performance. AI technologies are used to identify the components within EVBs, select the appropriate tools, determine which components should be disassembled, devise the sequence of operations, and test EVBs. Additionally, AI plays a key role in estimating the SoX and the RUL of EVBs, leveraging a variety of methods covered in earlier research. It is important to highlight that this review centers on robot-assisted processes for handling EVBs, omitting studies that focus exclusively on AI techniques for battery status determination.

A prevalent approach involves combining machine learning and AI, particularly in deep learning, for image analysis aimed at identifying and categorizing parts for disassembly. Most research in this field has focused on recognizing screw types and locations. Across seven different unscrewing tasks, AI was applied to detect and/or pinpoint screw heads, with the YOLO package being the most widely used tool. Additionally, OpenCV [119] was utilized in one method. AI's application extends beyond screws to the detection of connectors, BMS, mounting plates, and battery modules [16,71]. The focus on screws in EVB disassembly largely stems from the transfer of technologies developed for other applications, supported by existing datasets on various screw types. However, there is a pressing need to develop specialized datasets for categorizing EVB components to better identify parts specific to EVBs. Future work could involve the creation and distribution of public datasets tailored to classify EVB components.

AI and ML techniques have also been applied to analyze angle-torque curves, aiding in the detection of defective screws during disassembly [102]. Moreover, AI has been used to ensure the correct insertion of sockets into screw heads, through the development of an algorithm that employed linear force detection, as detailed in [61]. Additionally, AI plays a role in DSP, identifying and locating relevant components. Three studies utilized AI to plan unscrewing sequences, while one study applied AI to design the disassembly sequence for the entire battery pack.

5.3.1. Utilization of various types of robots

In the literature reviewed, articulated robots with six degrees of freedom were frequently employed. Numerous studies validated their methodologies through experimental setups or prototypes, often utilizing 3D-printed tools on collaborative robots with low payload capacities. Collaborative robots are popular in research due to their advanced safety features, which reduce the need for traditional safety barriers around the workspace. However, their drawbacks include lower payload limits and slower operational speeds, especially when compared to industrial robots. These limitations are particularly relevant in disassembly processes where larger, heavier components, such as battery modules weighing tens of kilograms, are involved. In such cases, robots with medium to high payload capacities are required.

Furthermore, the disassembly of certain components, such as sealed or bonded plates often found in modern EVB designs, demands forces that exceed the capacity of collaborative robots. While industrial robots are capable of handling these more demanding tasks, they also play a crucial role in minimizing the physical strain on operators by managing heavy or hazardous duties. Table 3 summarizes the various types of robots used across the reviewed studies.

In only two of the seventeen papers, the robots were capable of handling loads exceeding 15 kg, indicating a significant gap between research and practical industrial needs. Industrial-scale experiments were rarely reported, and the laboratory environments often lacked essential safety and fire prevention systems. Furthermore, the equipment used did not adhere to industrial standards, placing these solutions within TRLs 2 to 4.

5.3.2. Documented battery packs and available datasets

Table 4 presents a summary of publications offering detailed information on the disassembly of EVBs, categorized by vehicle make and model. These studies typically provide key elements such as disassembly sequence charts and bills of materials. Additionally, Table 5 extends this by listing not only EVB disassembly procedures but also includes links to datasets that can be used for training artificial intelligence models. Many of the articles indicate that supplementary datasets, beyond those provided, are available upon request.

6. Research directions

The following part examines possible methods for mechanising the remaining manual procedures in EVB disassembly, focussing on the second study question: What obstacles and possibilities emerge in using robotics to automate these activities?

6.1. Disconnecting electrical connectors and wire assemblies

The process of disconnecting wire assemblies and electrical connectors during the disassembly of an EVB is crucial for ensuring operational safety. This step is particularly important because many components, such as wiring assemblies, contactor-fuse units, and battery modules, carry high voltages that pose significant safety risks. These components remain electrically active, with the potential for dangerous voltage levels, until the wiring is properly disconnected. By removing these assemblies and connectors, technicians can safely handle and service the battery system without the risk of electrical shock or short

Table 3

Comparative analysis of robot models based on payload capacity and type.

Robot Make	Model	0–10 kg	11–50 kg	51–100 kg	101+ kg	Industrial	Collaborative	Ref.
ABB	IRB 140	✓				✓		[99]
ABB	IRB 4400			✓		✓		[71]
COMAU	NJ 220				✓	✓		[82]
Fanuc	LR-Mate 200iD/4S	✓				✓		[85]
Franka Emika	Panda	✓					✓	[110]
Kuka	KR 16		✓			✓		[102]
Kuka	KR 50		✓			✓		[82]
Kuka	LBR		✓				✓	[81]
Kuka	LWR	✓					✓	[17,98,109]
Techman	TM 14		✓				✓	[82,100]
UR	10e	✓					✓	[61,68,104,105]

Table 4

Detailed information of EVBs in reviewed publications.

Publication	Vehicle Make	Model	Battery Type	Year of Introduction	Motor Power (kW)	Range (km)	Charging Capability
[111]	BYD	Han EV	Lithium-Ion	2020	180 kW	605 km	Fast Charging, 100 kW DC
[82]	Fiat	500e	Lithium-Ion	2020	87 kW	320 km	85 kW DC Fast Charging
[110]	Nissan	Leaf 2011	Lithium Manganese Oxide (LMO)	2011	80 kW	160 km	50 kW CHAdeMO Fast Charging
[111]	Nissan	Leaf 2018	Nickel Manganese Cobalt (NMC)	2018	110 kW	270 km	50 kW CHAdeMO Fast Charging
[111]	BAIC BJEV	EU5	Lithium Iron Phosphate (LFP)	2018	160 kW	416 km	60 kW Fast Charging
[111]	Peugeot	e208	Lithium-Ion	2019	100 kW	340 km	100 kW DC Fast Charging
[75]	Tesla	Model S	Nickel Cobalt Aluminum (NCA)	2012	270 kW	600 km	Supercharging, up to 250 kW
[111]	Tesla	Model 3	Nickel Cobalt Aluminum (NCA)	2017	190 kW	500 km	Supercharging, up to 250 kW
[72]	Chevrolet	Bolt EV	Nickel-Rich Lithium-Ion	2017	150 kW	417 km	55 kW DC Fast Charging
[111]	Renault	Zoe R135	Lithium-Ion	2020	100 kW	395 km	50 kW DC CCS Fast Charging
[23]	Audi	e-tron 50	Lithium-Ion	2019	230 kW	336 km	120 kW DC Fast Charging
[17,70]	Audi	Q5 Hybrid	Lithium-Ion	2020	185 kW	50 km (electric only)	7.4 kW AC Charging

Table 5

Publications providing open access datasets for EVB research.

Reference	Dataset Description	Link to Dataset	Year of Publication
[16]	Dataset related to battery sorting and disassembly.	[120]	2021
[102]	Includes screw drive states during battery disassembly	[121]	2022
[106]	Point cloud data used for battery component segmentation	[122]	2021
[52]	Battery health metrics without degradation experiments	[123]	2020

circuits. In essence, this disconnection isolates the high-voltage elements, making it a critical procedure in the overall safety protocol for working on electric vehicle batteries. Failure to properly manage this disconnection could result in severe consequences, including injury or damage to the equipment.

This phase is particularly dangerous as the risk of electric shock persists until the wiring has been fully isolated. Once disconnected, the high voltage is confined to the battery modules, minimizing the risk to other components. Directly cutting the wires is not a feasible solution, as it may result in short circuits within the battery cells, potentially damaging both the cells and the wiring. These short circuits are a major contributor to thermal runaway, a significant safety concern in battery systems.

Despite the importance of this task, there is little research available on automating the disconnection of wire assemblies and connectors. The challenge stems from the flexible nature of wiring assemblies, which complicates their removal. Factors such as varying dimensions, orientations, and latching mechanisms of connectors, as well as the dense

arrangement and positioning behind other components, create additional difficulties. For instance, the Ford Transit EVB's wiring assembly is comprised of 14 distinct connectors, each equipped with a specific latching mechanism designed for secure connections. These connectors are not uniformly arranged; rather, some are positioned either vertically or horizontally behind other components within the system. This arrangement can complicate access to the connectors, as their placement behind other elements can create obstructions, making maintenance and modifications more challenging.

Automating the disassembly of wire assemblies is critical for enhancing safety in EVB dismantling processes. The key challenges include dexterous manipulation of locking mechanisms and handling flexible wires in confined spaces. The diversity in connector sizes and orientations requires specialized tools, while machine vision and AI could be integral to managing these tasks effectively. Fig. 6 illustrates a conceptual system that incorporates a gripper, ATC, and 3D camera to tackle these challenges.

Additionally, cognitive capabilities will be necessary to plan the tool's approach, particularly in adapting to random orientations and obstructions. Before disconnecting, ensure that the wire assembly does not obstruct access to the latching mechanism; if it does, reposition the assembly as necessary. Current disassembly technologies do not yet demonstrate the required autonomy for these tasks in industrial settings, underscoring the need for continued research and innovation.

6.2. Digital twins and battery passports

The concept of DTs for EVBs is closely tied to the proposed battery passports from industry leaders and the European Commission. Battery passports are linked to EVBs via identification tags, such as RFID or QR codes, that bridge the physical battery with its digital counterpart. This

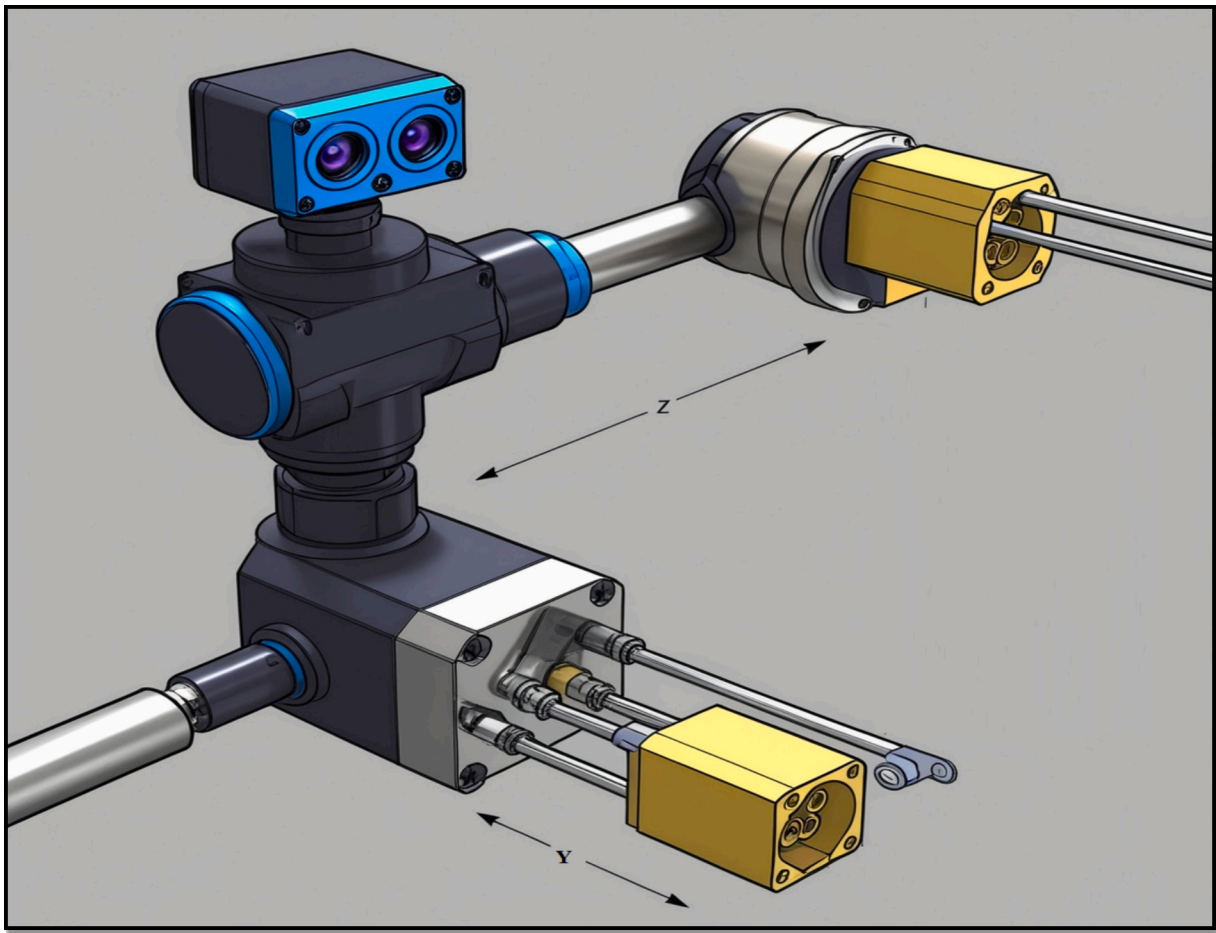


Fig. 6. Technical illustration of a custom gripper assembly for wiring connector removal.

connection allows efficient tracking for processes like recycling, remanufacturing, and repurposing. By incorporating the DT model, first introduced by Grieves for product lifecycle management, into battery passports, comprehensive testing, planning, and design for battery reuse and recycling can be realized. These processes are essential for supporting recycling efforts throughout the EVB lifecycle.

The DT framework provides manufacturers with crucial data regarding battery conditions, assisting decisions on whether the battery should be recycled, repurposed for energy storage in buildings, or reused as spare parts after the vehicle has been decommissioned. Furthermore, DTs can provide information on the chemical composition and amount of various materials in battery modules, facilitating recycling efforts.

Grieves emphasizes two primary reasons for updating the DT during the battery's disposal phase: improving future system designs and identifying hazardous materials for reduction or elimination in future iterations. As the electric vehicle market grows, encompassing cars, trucks, and vans, strategies for recycling and reusing batteries will be critical for minimizing the demand for raw material extraction for new battery production. Preut et al. identify key players in the EVB lifecycle, such as manufacturers, distributors, users, servicers, re-users, re-manufacturers, recyclers, logistics providers, and regulators. DTs continue to provide benefits for recyclers and re-users even after the EoL phase of the battery.

Fig. 7 provides a comprehensive illustration of the dynamic flow of information between the physical and digital twins throughout the entire lifecycle of an EVB. This continuous data exchange becomes especially significant when EVB modules are either repurposed for new functions or undergo remanufacturing. In such cases, valuable data from the modules' previous lifecycle, including their performance history,

condition, and usage patterns, are seamlessly transferred to the DT of the new product. For instance, if EVB modules are repurposed for energy storage systems in buildings, all relevant service data from their earlier use in electric vehicles is integrated into the DT of the repurposed energy storage system.

This integrated data not only ensures continuity but also enhances the accuracy of monitoring and optimization for the new application. Furthermore, digital twins play a pivotal role in the design phase of the repurposing process. They allow engineers to simulate various configurations, conduct performance testing, and explore alternative uses of the original product without physically altering the modules. This capability provides an opportunity to optimize the repurposed system for new environments, ensuring the most efficient use of the existing technology while reducing design and operational risks.

The battery passport represents a key innovation in promoting circular economy practices within the product lifecycle. In many cases, repurposing batteries offers a more efficient approach than traditional recycling methods, preserving energy and natural resources. The digital twin can extend beyond the physical disposal of the product, and the information gathered throughout the EVB's lifecycle—spanning design, production, maintenance, and disposal—can inform future designs, preventing past mistakes. Additionally, if sub-components are repurposed, their previous usage data can be transferred to the new product's DT. For EVBs, the BMS tracks key parameters, such as temperature, SoC, and charging/discharging cycles, ensuring safety and performance from assembly through the entire lifecycle.

This section also delves into the third research question, which seeks to examine the impact of forthcoming EU battery regulations and the proposed battery passport on the processes of EVB disassembly and



Fig. 7. Data requirements for the lifecycle stages of EVBs.

repurposing. Specifically, it aims to explore how these regulatory measures will reshape the future of battery management, particularly within the EU.

With the increasing emphasis on sustainability and environmental responsibility, the introduction of these regulations is expected to streamline the lifecycle management of EVBs, from their initial use to their eventual repurposing or disposal. The battery passport, a digital tool proposed by the EU, would facilitate more effective tracking of key information related to battery materials, usage history, and current health status. This transparency is anticipated to play a pivotal role in enhancing the efficiency of disassembly processes by providing critical insights into the battery's condition, thus enabling better-informed decisions regarding its reuse or recycling.

6.3. Improved adaptability and flexibility for addressing change

The process of disassembling EVBs is complex and requires the use of a range of specialized techniques and tools to achieve efficient and effective dismantling. A few studies in the literature have explored the use of ATC, which allows robotic systems to dynamically adjust and reconfigure themselves according to the specific tasks required during the disassembly process. This capability is particularly crucial given the vast diversity in battery designs, configurations, and assembly methods across different manufacturers and vehicle types. As such, the disassembly process cannot rely on a single, rigid method but must incorporate a broad array of strategies and approaches. This necessitates a high level of flexibility and adaptability in both the tools and the

techniques used to handle the various components, materials, and assembly configurations found in different EVB systems. The ability to quickly and effectively adapt disassembly processes is key to maximizing efficiency and minimizing the risk of damage to reusable or recyclable components.

To tackle this challenge, it's crucial to incorporate a variety of robotic tools, including multi-functional devices. For instance, tools like grippers integrated with cameras can help detect components, while systems for measuring voltage and resistance can assess the condition of battery modules. Complex operations, such as unscrewing, require seamless coordination between perception systems, external controllers, and the robotic mechanisms. By utilizing AI, the control over the disassembly process can be improved, integrating multiple perception sources. Fig. 8 shows how AI and multimodal inputs support tasks like unscrewing.

AI also plays a significant role in automating other disassembly tasks. For example, in the case of removing wiring connectors, input parameters like linear gripping force and the relative distance between gripper jaws are used instead of conventional metrics like rotation, torque, or speed. This enables the AI system to recognize when a connector has been successfully detached. Outputs such as speed and position are then adjusted, with the addition of a linear gripping force output to further refine the process. The tasks of unscrewing and detaching connectors differ in approach and input-output parameters. When detaching a connector, the system needs to prevent any contact with the wiring assembly, which moves away from the connector during the process.

These instances underscore the potential of AI to automate multiple facets of the EVB disassembly process. The outlined techniques could be

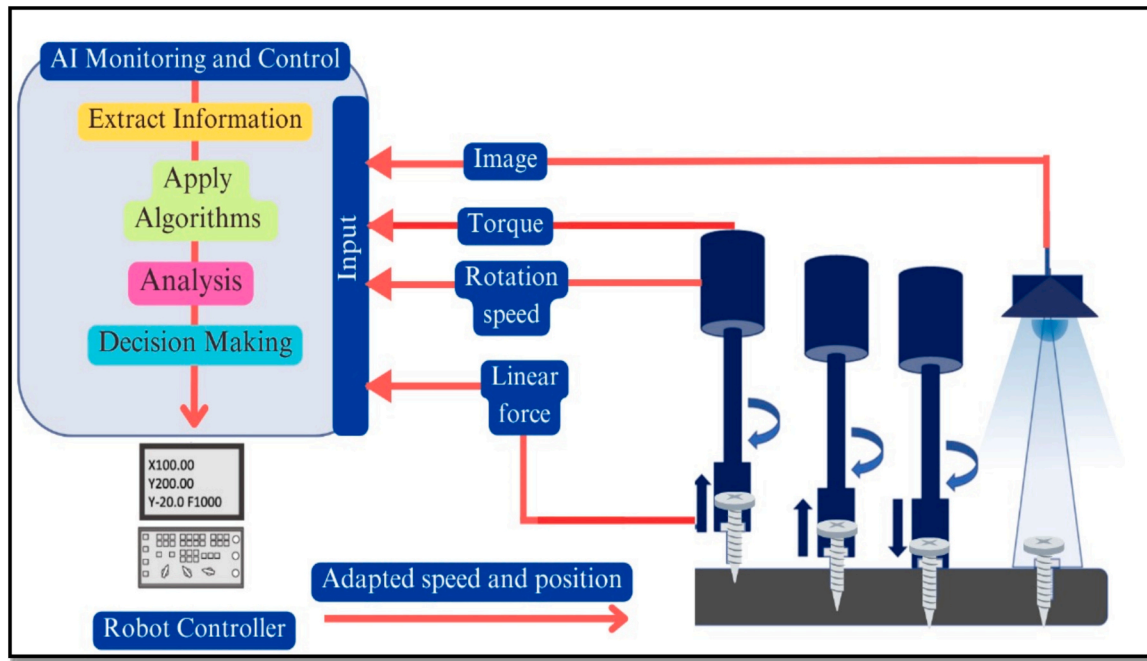


Fig. 8. Automated screw extraction: Identification, alignment, and loosening detection.

extended to optimize the remaining stages of disassembly, ultimately improving the system's flexibility and operational efficiency.

6.4. Teleoperation using alternative platforms

Teleoperation offers a valuable solution to enhance operator safety by allowing the control of the disassembly process from a remote location, away from the EVB processing area. This physical separation provides a secure environment for overseeing dismantling activities. However, a review of the literature indicates limited research advocating for the use of teleoperation in EVB disassembly, with only one study standing out. In other industries, such as nuclear decommissioning, teleoperation is commonly utilized for complex tasks.

The approach suggested in [126] proposes the use of head-mounted devices to deliver an AR interface and capture user movements for controlling remote manipulators during EVB disassembly. The system described includes a multimodal interface, enabling the operation of two remote manipulators through position tracking, eye and hand movements, and voice commands. This setup also supports collaborative efforts by allowing multiple users to interact in a shared virtual environment.

Integrating a more sophisticated XR interface alongside a DT of the disassembly cell could significantly improve teleoperation. As discussed in [127], trainees can remotely operate physical robots using digital twins, with the virtual environment representing fixed elements such as walls, workstations, and components necessary for training. This approach grants access to a virtual robotics laboratory, unrestricted by location or time, and allows digital twins to validate the trainees' paths to prevent collisions with static obstacles.

In the context of EVB processing, tele operators can use the digital twin to program and test procedures for disassembling new battery types, thereby enhancing the process. Additionally, the digital twin of the disassembly cell provides a safe training ground, ensuring that collision risks are minimized. Fig. 9 illustrates the application of a virtual factory setting designed to facilitate teleoperation and training in the disassembly of EVBs. This simulated environment allows operators to interact with virtual battery components, gaining hands-on experience in disassembly processes without physical constraints. By employing this virtual model, users can safely practice and refine their

skills, making the disassembly process more efficient and error-free in real-world applications. The virtual factory also supports remote operations, enabling teleoperated disassembly tasks, enhancing training effectiveness, and ensuring operator safety.

6.5. Enhancing safety protocols in battery disassembly

To maintain safety during the disassembly of EVBs, it is essential to work with only undamaged, intact batteries. Introducing automation in the pre-sorting phase can reduce the dangers posed by human proximity to potentially hazardous batteries. In recycling operations, damaged EVBs are often processed by crushing or melting the entire unit [34]. Research suggests that a combination of visual inspections and internal resistance assessments can be used to evaluate the safety of EVBs. Incorporating thermal imaging and gas detection systems can further boost safety by identifying potential gas leaks and instances of thermal runaway. These technologies assist in locating and isolating battery packs that may be hazardous or compromised.

In terms of HRC disassembly, deploying industrial robots equipped with the speed and separation monitoring (SSM) offers notable improvements in both productivity and safety. Nevertheless, despite recent progress in perception technologies, particularly safety-certified human-tracking sensors, these advancements have yet to be fully implemented in industrial environments.

Ensuring the safety of robotic disassembly systems for EVBs is a crucial challenge due to the presence of hazardous materials, high-voltage components, and the risk of thermal runaway. While automation offers significant advantages in minimizing human exposure to these risks, comprehensive safety measures must be integrated into robotic systems to ensure operational reliability.

6.5.1. Risk assessment in automated battery disassembly

Robotic disassembly introduces new safety concerns that must be addressed through systematic risk assessment methodologies. One key approach involves hazard identification (HAZID) and failure mode and effects analysis (FMEA), which assess potential risks associated with battery handling, cutting, and separation processes. AI-driven predictive maintenance can also be employed to monitor robotic components for early signs of wear or failure, preventing unexpected malfunctions that



Fig. 9. Simulated learning platform for EVB processing.

could compromise safety.

6.5.2. Fire and thermal runaway prevention

Thermal runaway remains a major concern when dealing with damaged or degraded EV batteries. To mitigate this risk, automated systems should incorporate real-time thermal imaging, gas detection sensors, and adaptive cooling mechanisms. For instance, machine learning algorithms can analyze battery surface temperature fluctuations and automatically adjust handling procedures to prevent overheating. Moreover, robotic disassembly lines can integrate enclosed, fire-resistant workstations with automated suppression systems to contain potential incidents.

6.5.3. Standardized safety protocols for HRC

In cases where full automation is not feasible, HRC plays a vital role in ensuring safety. Exoskeleton-assisted operators, force-limited cobots, and vision-based monitoring systems can enhance worker safety by providing real-time collision avoidance, ergonomic support, and adaptive task execution. Implementing safety-rated monitored stops (SRMS)

and SSM can further prevent accidents during manual interventions in robotic workspaces.

6.5.4. Best practices in secure handling and material recovery

Effective safety protocols must also consider the handling and transportation of disassembled battery components. Automated sorting and separation systems should employ robotic grippers with variable pressure controls to safely manipulate fragile battery cells without causing punctures or leaks. Additionally, the implementation of automated material classification and tracking systems can improve traceability, ensuring that recovered materials are processed in compliance with environmental and safety regulations.

6.6. Establishing a centralized EVB database

Research in this field has yielded substantial insights into twelve different types of EVBs. It is recommended to create a centralized repository that contains comprehensive profiles of all EVBs. This database should feature visual guides, disassembly protocols, technical details,

and datasets suitable for artificial intelligence applications. This resource would be especially beneficial for researchers and SMEs engaged in creating both manual and automated systems for EVB dismantling and sorting. Upon completion of the Battery Passport initiative, the gathered data could provide significant benefits to those involved in working with EVBs.

Access to comprehensive data, including battery dimensions, weight, and material composition, would significantly improve the logistics involved in transporting EVBs from dismantling locations to recycling centers. This would facilitate safe handling and contribute to the optimization of sorting and disassembly processes. Moreover, the technical insights offered by such a database would aid facilities, particularly within the OEM network, in effectively planning their production and disassembly activities. Additionally, this database would allow regulatory agencies to evaluate and compare various EVB designs, encouraging the creation of standardized EVBs that are easier to disassemble.

6.7. Progress in HRC

HRC has been successfully employed not only in the disassembly of EVBs but also in the breakdown of other electronic devices [128]. Techniques developed for such applications could be adapted to improve EVB disassembly processes. For example, references [129, 130] explore the disassembly of press-fitted components, using force sensors and active compliance to improve robot control. These methods allow robots to understand task execution while assisting human workers dealing with the same components. Future advancements in HRC could enhance the scalability of disassembly processes, thereby raising the overall LoA.

In reference [131], a comprehensive sensor-based platform is introduced, designed to streamline the disassembly process by generating sophisticated, high-level disassembly plans. This system takes into account the specific tools required at each step of the process, ensuring that the correct tools are selected and utilized efficiently. By coordinating the use of these tools across the various stages of disassembly, the platform optimizes the entire workflow. This integration not only improves precision in the disassembly process but also reduces potential delays or errors, as the system proactively plans and manages the tool usage based on the task at hand. The platform exemplifies how sensor technologies can be leveraged to automate complex disassembly tasks, contributing to more sustainable and effective production practices.

Similarly, reference [132] proposes an initial design for a general mechanical disassembly framework involving multiple task-specific workstations, where both robots and humans collaborate. This framework includes the extraction of disassembly features under operator supervision, while a cloud-based system manages data storage and schedules tasks for each workstation.

6.8. Technological feasibility of robotic disassembly in EVB processing

The disassembly of EVBs presents significant technological and economic challenges, necessitating advancements in robotic automation. While certain aspects of EVB disassembly can be readily automated, the high variability in battery pack designs, safety hazards, and operational complexities pose persistent challenges to full automation. Recent research has explored the feasibility of robotics, AI-driven machine vision, and HRC in EVB disassembly, highlighting both the potential benefits and current limitations.

Empirical studies suggest that robotic disassembly of EVBs holds significant potential for improving both economic viability and operational efficiency. Research indicates that approximately 57 % of pack-to-module (P2M) battery disassembly tasks can be fully automated, with an additional 24 % requiring only minimal human intervention. The integration of robotics and HRC has been shown to increase net annual revenue by 12.85 % while simultaneously reducing disassembly costs by 43.75 % [133]. However, complex disassembly tasks continue to pose

challenges due to the limitations in robotic dexterity and the need for enhanced AI-based decision-making. To address these issues, advances in ML, AI, and machine vision technologies have been leveraged to optimize robotic disassembly processes. AI-driven task planners have demonstrated the ability to streamline operations, achieving an average execution time of 34 s per step with a measurement error of less than 5 mm. Similarly, computer vision systems incorporating 2D and 3D imaging have improved the accuracy of localizing fasteners and threaded connectors, thereby enhancing operational precision. Despite these advancements, AI models still face difficulties in real-time adaptation to varying battery pack configurations, necessitating further refinement through training on diverse datasets [134].

Additionally, the development of specialized robotic tools has contributed to the feasibility of automation in EVB disassembly. Robotic cutting tools, for instance, offer superior precision over manual methods, with cutting speed and direction playing a crucial role in ensuring high-quality results [135]. Custom gripping mechanisms have also been designed for robotic systems, although they continue to struggle with dexterous pick-and-place tasks, an area where human flexibility remains unmatched [136]. Fully automated unscrewing mechanisms have been introduced to enable robots to handle fasteners autonomously even under positional uncertainties, thereby improving scalability and safety in industrial applications [137].

Despite these technological strides, full autonomy in EVB disassembly remains infeasible due to the inherent complexity and unpredictability of the process. Consequently, human-robot collaboration has emerged as a practical alternative, integrating sensor-based robotics, cost-effective tooling, and real-time safety monitoring to establish a more ergonomic and scalable workspace [138]. Given the potential risks associated with explosive atmospheres during battery disassembly, HRC ensures safer operations by enabling human operators to oversee high-risk tasks while robots perform repetitive and hazardous activities [139]. However, the high investment costs of sensor-integrated collaborative robots present a financial barrier, necessitating further reductions in implementation costs to facilitate widespread industrial adoption [140]. Thus, while robotic and AI-driven automation significantly enhance the efficiency and safety of EVB disassembly, ongoing research and development are essential to address existing limitations, improve real-time adaptability, and optimize cost-effectiveness for large-scale deployment. Table 6 below provides a comprehensive summary of the automation potential, technological advancements, and challenges in robotic EVB disassembly based on recent studies.

6.9. Regulatory and industry collaboration for standardized EVB disassembly

The effective implementation of robotic disassembly for EVBs necessitates a structured collaboration between policymakers, robotics engineers, and battery manufacturers to establish standardized frameworks for automation, safety, and sustainability. Currently, the lack of universal battery design standards poses a major challenge for automation, requiring robotic systems to handle a wide range of battery chemistries, architectures, and assembly techniques. To facilitate large-scale automation, harmonized regulations should focus on DfD principles, ensuring that EVBs are engineered for ease of disassembly and material recovery.

6.9.1. Practical enforcement of regulations

The enforcement of battery disassembly regulations can be approached through a combination of policy mandates, industry incentives, and technological integration. The following strategies can enhance regulatory enforcement:

- **Mandating DfD Requirements:** Governments should establish clear DfD guidelines, such as the use of standardized fastening mechanisms (e.

Table 6
Comparative analysis of robotic disassembly feasibility in EVB processing.

Study	Focus Area	Key Findings	Limitations & Challenges
[133]	Economic Benefits of Automation	57 % of pack-to-module (P2M) battery disassembly tasks are fully automatable; an additional 24 % require minimal human intervention. Automation increases net annual revenue by 12.85 % and reduces disassembly costs by 43.75 %.	Complex disassembly tasks, such as adhesive removal and non-standard battery pack configurations, remain difficult to automate due to high variability.
[134]	AI Task Planning for Disassembly	AI-driven disassembly planners optimize robotic movement, achieving an average task execution time of 34 s per step with a measurement error below 5 mm.	AI models require large-scale training datasets and high computational power to adapt to real-time changes in battery configurations, limiting efficiency in unpredictable environments.
[135]	AI-Driven Robotic Cutting Tools	AI-assisted robotic cutting achieves 98 % precision, outperforming manual methods in speed and consistency. Automation reduces human errors and increases the lifespan of reusable battery components.	AI-driven tools require extensive training to adapt to the diverse geometries and materials of different battery designs, limiting their generalizability.
[136]	Robotic Gripping & Cutting Tools	Custom robotic end effectors, including force-adaptive grippers and high-precision cutting tools, enhance efficiency by 30 % over manual disassembly. Cutting tools ensure ± 1 mm accuracy in incision depth.	Grippers face challenges in dexterous pick-and-place operations, particularly with irregularly shaped battery modules and delicate components. Cutting tools struggle with non-uniform materials.
[137]	Computer Vision for Automation	2D and 3D imaging technologies improve fastener and connector localization accuracy by 85 %, reducing manual intervention in screw removal by 40 %.	Computer vision struggles with occlusions, reflective surfaces, and non-standardized battery pack layouts, requiring human verification for uncertainties.
[138]	Sensor-Based HRC for Disassembly	Sensor-integrated collaborative robots enhance safety by enabling real-time monitoring and adaptive force control, ensuring compliance with ergonomic standards. Scalable for industrial applications.	High implementation costs, estimated at \$250,000 per workstation, pose a significant barrier to widespread industrial adoption. Further cost reductions are necessary for feasibility.
[139]	HRC in EVB Disassembly	HRC-based systems improve disassembly flexibility, achieving 20 % faster processing times while maintaining productivity and human oversight. Robots handle repetitive tasks, reducing worker fatigue by 35 %.	High risk of thermal runaway and explosive atmospheres during battery disassembly necessitates strict safety measures. Human intervention is still required for real-time risk mitigation.
[140]	Feasibility of Full Automation	Automation improves efficiency and safety in disassembly, reducing manual labor by 60 %. Essential for sustainable EV battery recycling and lifecycle management.	Full automation remains infeasible due to the complexity, high uncertainty, and varying designs of battery packs, requiring hybrid automation approaches.

g., screws instead of adhesives) and modular battery pack designs that facilitate robotic disassembly.

- **Digital Twin and Battery Passports:** Implementing digital twin technology—a virtual representation of each battery—can provide real-time data on battery composition, structural layout, and safe disassembly procedures. This will enable AI-driven robotic automation and reduce safety risks.
- **Extended Producer Responsibility (EPR) Programs:** Enforcing EPR policies will make manufacturers responsible for the entire lifecycle of EV batteries, incentivizing them to adopt designs that are compatible with robotic disassembly.
- **Regulatory Incentives for Automation-Ready Designs:** Governments can provide financial incentives (tax credits, subsidies) for manufacturers that develop robotic-friendly battery designs, promoting industry-wide standardization.

6.9.2. Industry collaboration for standardized frameworks

The development of standardized frameworks for robotic battery disassembly requires active collaboration between three primary stakeholders:

- **Policymakers** – Develop and enforce regulations that promote automation-friendly battery designs.
- **Robotics Engineers** – Design robotic systems capable of adapting to various battery architectures while ensuring safety and efficiency.
- **Battery Manufacturers** – Modify battery designs to enhance compatibility with automated disassembly systems.

To illustrate these roles, Table 7 outlines the key responsibilities of each stakeholder in establishing an effective regulatory framework for EV battery disassembly.

6.9.3. Conceptual framework for a standardized robotic disassembly system

A standardized robotic disassembly system should integrate:

- A digital twin database containing real-time battery information.
- AI-driven machine vision to identify different battery types and optimize disassembly steps.
- Automated modular separation for efficient material recovery.
- Regulatory compliance monitoring, ensuring each step adheres to safety and sustainability regulations.

6.9.4. Future directions and policy recommendations

To ensure the long-term success of robotic EVB disassembly, future research and policy should focus on:

- Developing internationally recognized standards for battery design and disassembly automation.
- Strengthening collaboration between governments, automotive industries, and AI researchers to enhance robotics integration.
- Implementing real-time compliance monitoring systems to ensure safety and efficiency in robotic disassembly processes.

By fostering strong collaboration among policymakers, robotics engineers, and manufacturers, the industry can transition toward a fully automated, safe, and environmentally sustainable EV battery disassembly system.

7. Conclusion

This study has provided an in-depth and structured examination of robotic solutions for processing EVBs within our institution. We began by evaluating the critical need for automation and the development of robotic systems specifically tailored to EVB handling. Our analysis included the formulation of a standard disassembly sequence,

Table 7

Roles and contributions of key stakeholders in standardizing robotic EVB disassembly.

Stakeholder	Key Responsibilities	Proposed Actions	Expected Benefits
Policymakers	- Develop global regulations for standardizing battery disassembly. - Implement EPR policies to ensure manufacturers participate in recycling initiatives. - Promote the adoption of Digital Battery Passports for tracking battery lifecycle.	- Establish incentives (tax credits, R&D funding) for manufacturers adopting DfD principles. - Introduce AI-driven regulatory monitoring to ensure compliance in battery disassembly. - Mandate standardized labeling for battery composition, safety ratings, and disassembly steps.	- Encourages eco-friendly battery designs aligned with automation needs. - Reduces hazards in disassembly by ensuring transparency in battery structure. - Ensures accountability among manufacturers, improving material recovery rates.
Robotics Engineers	- Develop AI-powered robotic systems capable of handling various battery types. - Improve machine vision for accurate battery identification and separation. - Optimize robotic end-effectors for safe, precise disassembly.	- Design adaptive robotics capable of adjusting to different battery models. - Implement predictive maintenance algorithms to enhance robotic efficiency. - Develop human-robot collaboration frameworks for handling complex disassembly tasks.	- Increases efficiency and accuracy in automated disassembly. - Minimizes human exposure to hazardous battery components. - Enhances scalability of battery recycling operations.
Battery Manufacturers	- Standardize battery architectures to facilitate robotic automation. - Incorporate DfD principles into battery pack designs (e.g., modular packs, minimal adhesives). - Provide real-time data access through Digital Battery Passports.	- Replace adhesives with mechanical fasteners (screws, clips) for easier disassembly. - Integrate QR codes, RFID tags, and serial numbers for instant robotic identification. - Collaborate with robotics engineers to develop automation-compatible battery structures.	- Reduces disassembly complexity and costs. - Improves battery reusability and recyclability. - Strengthens supply chain sustainability by enhancing material recovery.

identifying safety concerns, and discussing the lack of standardized designs and regulations within the industry.

In continuation, we examined the available literature on different aspects of EVB processing. This review included discussions on testing methods, the use of machine vision and machine learning technologies, robotic programming, disassembly task planning, human-robot interaction and control systems, as well as safety measures. Moreover, we addressed the design aspects of EVBs that could support more efficient robotic processing, while also citing relevant research in these areas.

As the number of end-of-life EVBs is expected to rise significantly over the coming decade, there is a pressing need to develop more advanced technologies for repurposing, recycling, and remanufacturing these batteries. Our research reveals a substantial gap in the development of sophisticated robotic tools and processes for EVB handling. Currently, a significant portion of the disassembly process is carried out manually, with automation tools still in their early stages of development, emphasizing the necessity for future advancements.

In terms of HRC, the field remains underexplored. The current designs and architectures of EVBs often require human involvement, a limitation that is partially attributable to the constraints of contemporary artificial intelligence. To improve the efficiency of robotic and automated systems in this field, the industry would gain from the implementation of stricter regulations and design standards that push manufacturers to develop EVBs that are more compatible with automated processes.

CRedit authorship contribution statement

Vivek Saxena: Writing – review & editing, Writing – original draft, Visualization, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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